



The University of Jordan

**School of Engineering
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Lightweight Bilingual Image Captioning Model

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Submitted in partial fulfillment of the requirements of B.Sc. Degree in Computer Engineering

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SUPERVISOR CERTIFICATION

I hereby certify that the students in this project have **successfully finished** their senior year project and by submitting this report they have fulfilled in partial the requirements of B.Sc. Degree in Computer Engineering.

☐

I hereby certify that the students in this project have not completed their senior year graduation project and **I do not approve** that they proceed to the discussion.

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I suspect that the students have **violated** one or more of the clauses in the **ethical statement** and I suggest that an investigation committee look into the matter.

☐

Dr. Fahed Jubair

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DEDICATION

I would like to dedicate this effort to my family, whose constant support, encouragement, and patience have been invaluable throughout this journey. Their belief in me has been a source of strength and motivation during the most challenging moments.

A special thanks goes to **Dr. Fahed Jubair** for his continuous guidance, valuable feedback, and support during the course of this project. I am also deeply grateful to all my professors who have contributed to my learning and growth over the past years. Their dedication and knowledge have left a lasting impact on my academic and professional development.

SYMBOLS, ABBREVIATIONS, AND ACRONYMS

CLIP	Contrastive Language–Image Pre-training
ViT	Vision Transformer
mT5	Multilingual Text-To-Text Transfer Transformer
LoRA	Low-Rank Adaptation
CLS	Classification Token
token	
NMT	Neural Machine Translation
LSTM	Long Short-Term Memory

ABSTRACT

In this project, I developed a bilingual image captioning system that generates meaningful and context-aware descriptions for images in both English and Arabic. Image captioning plays an important role in connecting computer vision with natural language processing, allowing systems to describe visual content using human language. My goal was to design a model that not only produces fluent and accurate captions but also supports multilingual communication, especially for less-resourced languages such as Arabic.

To achieve this, I combined a visual representation module with a multilingual language generation model capable of adapting to both English and Arabic inputs. I trained the system on a bilingual dataset and used language-specific tokens to guide the model in generating captions in the desired language. I evaluated the model based on its ability to generate fluent, relevant, and complete captions across both languages. The results showed that the system achieved a strong balance between the two languages and demonstrated reliable generalization.

Through this work, I aimed to contribute to the growing demand for inclusive AI technologies that can serve diverse user communities. The system I developed can be applied in various real-world contexts, including education, accessibility tools, and content management platforms. This project also highlights the importance of supporting linguistic diversity in AI development and provides a foundation for future research in multilingual vision-language systems.

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CHAPTER 1

INTRODUCTION

Image captioning is a key task in artificial intelligence (AI) that combines computer vision and natural language processing (NLP) to describe images using natural language. It plays an important role in connecting visual understanding with language, supporting applications like helping visually impaired users, improving image search, and making human–AI interaction more intuitive.

In this project, we developed a bilingual image captioning system that can generate captions in both English and Arabic. Our model uses CLIP with a Vision Transformer (ViT) to extract image features, and the multilingual mT5 model to generate accurate text descriptions. By including language control tokens, the system can switch between Arabic and English easily, making it accessible to a wider, multilingual audience.

1.1. Problem Definition

With the increasing amount of image-based content shared across digital platforms, it has become important to develop systems that can generate meaningful descriptions automatically. Manual captioning is time-consuming and not scalable, while many existing models rely on object detection techniques that often miss the full context of the image.

Another key issue is that most available captioning models are focused only on the English language. This creates a barrier for users who speak other languages, especially Arabic, which is underrepresented in many vision-language tasks. Our project aims to solve this by designing a bilingual image captioning model that supports both Arabic and English, making it more inclusive and useful for a wider audience.

1.2. Proposed Solution

To tackle the challenges mentioned earlier, we developed a bilingual image captioning system that supports both English and Arabic. The system uses a modular deep learning architecture made up of five main parts:

- **Feature Extraction:** We use a pre-trained CLIP model with a Vision Transformer (ViT) backbone to get meaningful visual features from images. CLIP is trained on large datasets of image-text pairs using contrastive learning, which helps it learn the connection between images and text.
- **Sequence Generation:** The visual features extracted by CLIP are projected into the embedding space expected by the mT5 model. mT5 is a multilingual text-to-text transformer that can generate fluent captions in several languages, including English and Arabic.
- **Parameter-Efficient Fine-Tuning with LoRA:** To make training more efficient and reduce memory usage, we apply Low-Rank Adaptation (LoRA) to only a small number of parameters in the mT5 model. This lets us fine-tune the model on limited hardware while maintaining good performance.

- **Bilingual Language Control:** The model uses special tokens, <ENGLISH> and <ARABIC>, added at the beginning of each caption during training and inference. This guides the model to generate captions in the requested language.
- **Evaluation Strategy:** We designed a simple scoring method to evaluate the captions. Our criteria include correct identification of objects and actions, grammatical fluency, and whether the caption matches the target language.

This approach balances model performance and training efficiency while supporting two languages with a single system.

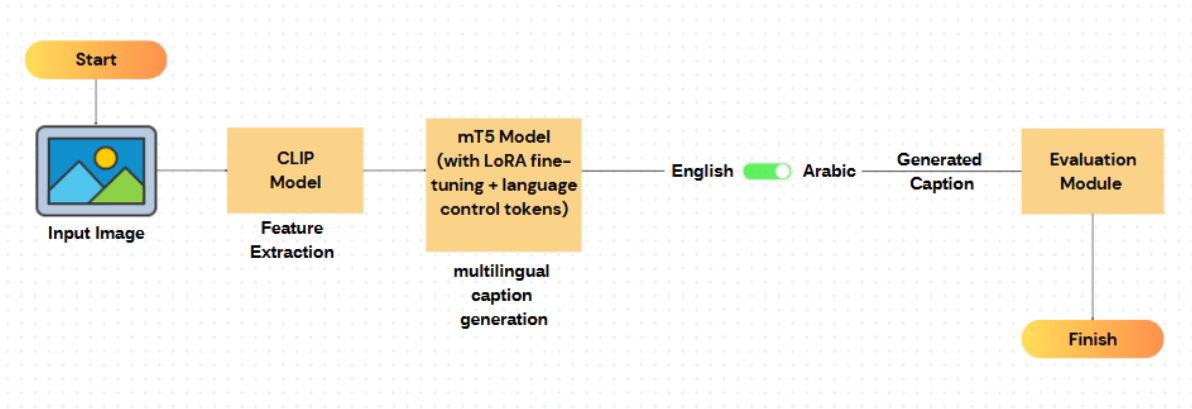


Figure 1: Overview of the Bilingual Image Captioning System Architecture

The architecture of the proposed bilingual image captioning system is illustrated in Fig.1. The system begins by extracting visual features from an input image using a pre-trained CLIP model with a Vision Transformer (ViT) backbone. These features are then projected into the embedding space of the mT5 model, a multilingual text-to-text transformer. To enable efficient training, Low-Rank Adaptation (LoRA) is applied to selectively fine-tune mT5 parameters. Caption generation is guided by language control tokens (“<ENGLISH>” or “<ARABIC>”), allowing the system to produce output in the desired language. Finally, an evaluation module assesses the quality of generated captions based on predefined criteria.

1.3. Project Deliverables

The final outcome of this project consists of several key components. First, we provide a trained deep learning model that combines CLIP-ViT for visual feature extraction with mT5 for multilingual text generation. This model is capable of producing image captions in both English and Arabic.

To facilitate practical usage and demonstration, a user-friendly interface has been developed. It allows users to upload images, select the desired language, and view the generated captions in real time. This interface serves as both a testing tool and a visualization platform.

The system is trained and evaluated using a bilingual dataset, including the Flickr30k dataset, which provides 31,702 images for training, and a custom-built test set of 200 images curated for bilingual evaluation. This combination ensures a diverse and representative training environment.

For assessing the model’s performance, we include a detailed evaluation framework. It incorporates predefined scoring metrics that measure the readability of the generated text,

accuracy in object and action identification, and the completeness of the captions relative to the image content.

1.4. Project Impact

This project goes beyond technical achievement by providing meaningful benefits in both economic and social areas. By generating bilingual image captions automatically, the system offers practical solutions that can improve content accessibility, reduce manual work, and support multilingual technology development.

1.4.1. Economic Impact

The system helps reduce the time and cost of manual image captioning and translation by automatically generating context-aware captions in English and Arabic. This is especially useful in industries like:

- Digital media – for faster content management and multilingual publishing
- E-commerce – for better product tagging and search optimization
- Archiving and documentation – for automated image labeling and classification

By supporting two widely used languages, the system also increases content accessibility and helps businesses reach a broader audience, enabling cross-language engagement and expanding their market reach.

1.4.2. Social Impact

From a social perspective, the project promotes accessibility, inclusivity, and education:

- It generates bilingual captions that are compatible with screen readers, helping visually impaired users understand image content.
- It contributes to multilingual AI development, especially by supporting Arabic, a language often underrepresented in AI tools.
- The project can be used in education and research, offering:
 - A reproducible model architecture
 - A bilingual dataset
 - Evaluation tools for academic and development purposes
- It can also support language learning by helping students see how the same image is described in both English and Arabic, making it a useful tool in bilingual education.

CHAPTER 2

RELATED WORK

Image captioning has evolved significantly, starting from simple encoder-decoder models to more advanced vision-language transformers. While early models focused mostly on English, recent works explore multilingual and bilingual captioning. Below are some key approaches and their strengths and limitations, especially in the context of supporting Arabic alongside English.

a. Show and Tell Architecture

The "Show and Tell" model by Vinyals et al. was one of the earliest successful image captioning models [1]. It used a Convolutional Neural Network (CNN) to extract visual features and passed them to a Recurrent Neural Network (RNN), typically an LSTM, to generate captions word-by-word.

Why it was important:

- Established the encoder-decoder approach for captioning.
- Achieved strong results in English on datasets like MS COCO.

Limitations:

- No attention mechanism, so it struggled with complex scenes.
- Only supported English, which made it unsuitable for multilingual tasks.

Despite its limits, this model laid the foundation for modern image captioning systems.

b. Trilingual Captioning using mT5 or MarianMT

This method involves generating captions in English and translating them to Arabic using models like MarianMT or mT5 [2]. It's often used when bilingual datasets are unavailable.

Why it works:

- Easy to implement using pre-trained English captioning and translation models.
- Highly modular—you can switch or fine-tune components independently.

Challenges:

- Translation may sound unnatural or lose cultural nuances.
- Image is not used during translation, so visual context is missed.
- Slower due to the two-step process.

Even with these drawbacks, it remains a practical solution for resource-limited projects.

c. Multilingual ClipCap

Multilingual ClipCap is an extension of ClipCap that supports captioning in languages like Arabic by combining CLIP embeddings with multilingual decoders such as mT5 or mBART [3].

Why it's effective:

- Doesn't require large multilingual datasets.
- Lightweight: trains only a small projection layer to adapt CLIP features.
- Supports switching languages by changing tokens.

Limitations:

- Output quality relies heavily on how well the decoder handles Arabic.
- With limited Arabic training data, captions may lack fluency or style.

This modular and efficient approach influenced the architecture of our own system.

d. Multilingual ClipCap

This method involves generating captions in English and translating them to Arabic using an NMT model [4]. It's a practical solution for adding bilingual support without having to train a multilingual captioning model from scratch.

Why it works:

- Reuses strong English captioning models and existing translation tools.
- Saves time and resources when bilingual datasets are limited.

Trade-offs:

- Translations may lose context or sound unnatural.
- The translation model can't adjust captions based on image content.
- Processing time increases slightly due to the dual-step flow.

Despite these limitations, this method is effective for many applications like accessibility, language learning, or multilingual web platforms.

CHAPTER 3

SOLUTION DESCRIPTION AND IMPLEMENTATION

3.1. Preliminary Evaluation of Previously Used Vision-Language Models

Before finalizing the architecture for the bilingual image captioning model, I explored multiple pretrained vision-language models to assess their suitability for generating captions in both Arabic and English. Two major approaches were evaluated: Florence, developed by Microsoft, and Vision Transformer (ViT) in its standard and CLIP-enhanced forms.

a. Evaluation of Florence for Bilingual Captioning

One of the first models I experimented with was Florence [5], a large-scale vision-language model built by Microsoft. It was designed primarily for general-purpose visual understanding in English, and its architecture adopts a causal decoder-only structure, which is effective for generating fluent and semantically rich English captions. Florence integrates both visual input and textual output, making it suitable for tasks like captioning, and is compatible with LoRA fine-tuning—making it attractive for limited-resource environments like Google Colab.

For experimentation, I fine-tuned the microsoft/Florence-2-base model using the Flickr8k dataset that included captions in both Arabic and English. The training was conducted on Google Colab with a 12 GB GPU, and due to memory constraints, only one epoch was completed. LoRA was applied to the attention modules (q_proj and v_proj) to reduce resource usage, and the model's processor was used to tokenize the bilingual captions.

However, while Florence performed reasonably well in English caption generation, it struggled significantly with Arabic:

- It was mainly trained on English data, leading to language bias.
- The tokenizer could not handle Arabic morphology and script well, producing fragmented or incorrect outputs.
- The model's design was monolingual, optimized for English text generation and lacked true multilingual capabilities.

These challenges made Florence unsuitable for bilingual tasks without extensive retraining on a diverse multilingual dataset.

b. From Vision Transformer to CLIP-ViT: Enhancing Visual-Language Alignment

I also evaluated the use of Vision Transformers (ViT) as standalone image encoders due to their strong performance in visual understanding tasks [6]. ViT introduced a

novel approach that processes images by dividing them into fixed-size patches (e.g., 16×16), treating them like word tokens, and feeding them into a transformer network. This method, combined with self-attention, enables ViT to capture long-range dependencies within an image and achieve strong performance when trained on large datasets such as ImageNet-21k or JFT-300M.

However, standard ViT models are not trained with text data and lack direct alignment with natural language, making them insufficient for image captioning unless paired with an external language model and a bridging mechanism.

To address this, I shifted to CLIP-ViT, which builds upon the ViT architecture but is trained using Contrastive Language–Image Pretraining (CLIP). Developed by OpenAI, CLIP was trained on 400 million image–text pairs and embeds both image and text into a shared semantic space. This multimodal training allows CLIP-ViT to serve as a much better foundation for tasks like captioning, especially when paired with a multilingual decoder.

The advantages of CLIP-ViT over standard ViT are summarized below:

Table 1 - Comparison between CLIP-ViT and Standard ViT for Image Captioning Tasks

Feature	CLIP-ViT	Standard ViT
Multimodal Training	Trained jointly with text using contrastive learning	Trained only for image classification
Semantic Alignment	Directly aligns visual features with text embeddings	Requires external alignment mechanisms
Zero-Shot Capabilities	Strong — generalizes well to unseen tasks	Limited to specific pretraining tasks
Pretrained on Image-Text Pairs	Yes (e.g., 400M pairs)	No (e.g., ImageNet-only)
Compatibility with Multilingual Decoders	High — works well with decoders like mT5	Requires additional mapping or adaptation

By switching to CLIP-ViT, paired with mT5 as the decoder, I was able to:

- Leverage semantically meaningful pretrained visual embeddings.
- Use precomputed features, which reduced training time.

- Generate captions in both Arabic and English using a decoder trained on over 100 languages.

This architecture resolved many of the issues seen with Florence and standard ViT. Florence was limited by its English-centric training, and ViT lacked language understanding. CLIP-ViT, however, provided the necessary bridge between vision and language, enabling better performance for bilingual captioning tasks.

3.2. Bilingual Image Captioning Model

In my project, the image captioning model is hosted on a server, and users can interact with it through a web-based interface built using Gradio. The interface allows users to upload an image and select their preferred language—Arabic or English. Once an image is submitted, the system generates a relevant caption in the chosen language. This setup makes the tool easy to use and accessible online, without requiring users to install any software or go through complicated setup steps.

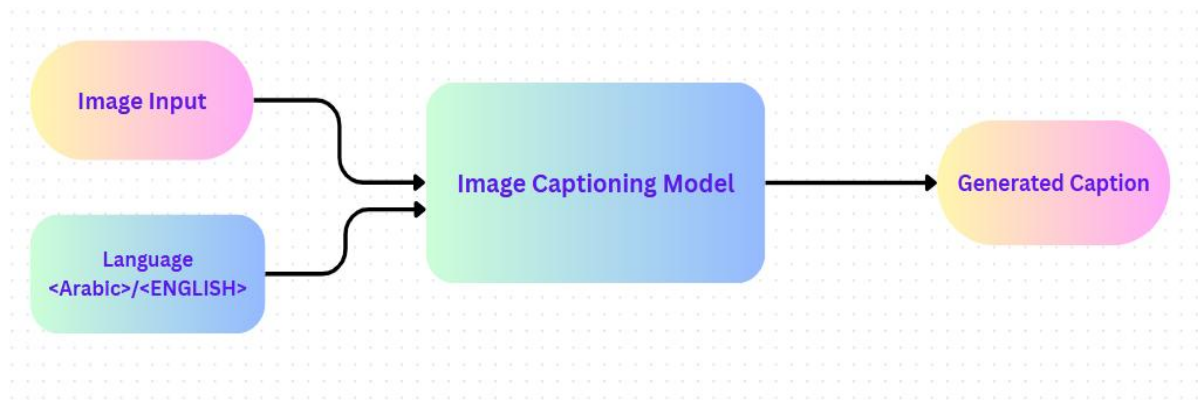


Figure 2 - Overview of the Bilingual Image Captioning Model Workflow

3.2.1. Data Collection

To develop and evaluate our bilingual image captioning model, we followed a two-phase dataset approach. During the initial experimentation phase, we used the **Flickr8k** dataset, which includes 8,000 images, each paired with five different English captions. These captions provide a variety of descriptions for the same image, helping the model learn from multiple perspectives. This dataset was especially useful for training lightweight prototypes and comparing the performance of different model settings in a manageable and efficient way.

a. Phase 1: Preliminary Training and Evaluation using Flickr8k

In the initial phase of the project, we used the Flickr8k dataset to test, evaluate, and identify the most suitable architecture for bilingual image captioning. The dataset contains 8,000 images, each with five descriptive English captions, resulting in a total of 40,000 English captions. These multiple captions offer diverse linguistic perspectives for each image, helping the model learn to generate more accurate and generalizable descriptions.

To enable bilingual training and evaluation, we extended the dataset by incorporating an Arabic version of the Flickr8k captions. This bilingual dataset consists of two parts:

the original English captions and a corresponding set of Arabic captions. While the English version provides five captions per image, the Arabic version includes three human-translated captions per image. Despite the smaller number of Arabic captions, they are directly aligned with the images, which ensures meaningful bilingual learning. This combined dataset was essential for benchmarking the model’s performance and fine-tuning different configurations during the early stages of development.

1. Sample from Flickr8k English:

A	B	C
name	caption	language
1000268201_693b08cb0e.jpg	A child in a pink dress is climbing up a set of stairs in an entry way .	<ENGLISH>
1000268201_693b08cb0e.jpg	A girl going into a wooden building .	<ENGLISH>
1000268201_693b08cb0e.jpg	A little girl climbing into a wooden playhouse .	<ENGLISH>
1000268201_693b08cb0e.jpg	A little girl climbing the stairs to her playhouse .	<ENGLISH>
1000268201_693b08cb0e.jpg	A little girl in a pink dress going into a wooden cabin .	<ENGLISH>
1001773457_577c3a7d70.jpg	A black dog and a spotted dog are fighting	<ENGLISH>
1001773457_577c3a7d70.jpg	A black dog and a tri-colored dog playing with each other on the road .	<ENGLISH>
1001773457_577c3a7d70.jpg	A black dog and a white dog with brown spots are staring at each other in the street .	<ENGLISH>
1001773457_577c3a7d70.jpg	Two dogs of different breeds looking at each other on the road .	<ENGLISH>
1001773457_577c3a7d70.jpg	Two dogs on pavement moving toward each other .	<ENGLISH>

Figure 3 - English Captions for each Image

2. Sample from Flickr8k Arabic:

A	B	C
name	caption	language
1000268201_693b08cb0e.jpg	طفلة صغيرة تتسلق إلى مسرح خشبي	<ARABIC>
1000268201_693b08cb0e.jpg	طفلة صغيرة تتسلق الدرج إلى منزلها	<ARABIC>
1000268201_693b08cb0e.jpg	فتاة صغيرة في ثوب وردي تذهب إلى المقصورة الخشبية	<ARABIC>
1001773457_577c3a7d70.jpg	كلب أسود وكلب ثلاثي الألوان يلعبان مع بعضهما البعض على الطريق	<ARABIC>
1001773457_577c3a7d70.jpg	كلب أسود وكلب أبيض ببقع بنية يحدقان في بعضهما البعض في الشارع	<ARABIC>
1001773457_577c3a7d70.jpg	كلاب على الرصيف تتحرك نحو بعضها البعض	<ARABIC>
1002674143_1b742ab4b8.jpg	فتاة صغيرة مغطاة بالطلاء تجلس أمام قوس قزح	<ARABIC>
1002674143_1b742ab4b8.jpg	فتاة صغيرة تجلس أمام قوس قزح ملون كبير	<ARABIC>
1002674143_1b742ab4b8.jpg	فتاة أمام لوحة قوس قزح	<ARABIC>

Figure 4 - Arabic Captions for each Image

This dataset gave us the opportunity to experiment with different encoder-decoder architectures and assess the quality of the generated captions in both English and Arabic. Based on the results, we were able to compare various model configurations and identify their strengths and limitations. These insights ultimately led us to expand the training process using a larger and more diverse dataset to improve overall performance.

b. Phase 2: Building a Bilingual Training Set from Flickr30k

After achieving promising results during the initial experiments with Flickr8k, we moved on to using the more comprehensive Flickr30k dataset to further improve our model’s performance. This dataset contains 31,702 unique images with precomputed image embeddings, and each image is annotated with five English captions and five corresponding Arabic translations. In total, the dataset includes 157,645 captions in English and 157,645 captions in Arabic, making it a well-balanced and bilingual resource. The rich and diverse annotations in both languages provided a strong

foundation for training our model to generate accurate and meaningful captions in both English and Arabic.

Table 2 - Dataset Statistics Summary

Statistic	Value
Number of unique images with embeddings	31,702
Number of English captions	157,645
Number of Arabic captions	157,645

To prepare the dataset for bilingual caption generation in both English and Arabic, we extended the original captions by translating them into Arabic using the GPT-4o model through the OpenAI API. This method provided accurate, context-aware translations while maintaining the meaning and richness of the original English captions.

A custom Python script was developed to automate the translation workflow. The script processed captions in batches, sending them to the OpenAI API and retrieving high-quality Arabic translations. For bilingual training purposes, each translated caption was aligned with its corresponding English version. Additionally, each caption was tagged with language-specific tokens, namely <ENGLISH> and <ARABIC>, to explicitly indicate the language. This structured labeling enabled the model to distinguish between languages during training, thereby enhancing its bilingual captioning capability.

```
import openai
import time

# Set your OpenAI API key
client = openai.OpenAI(api_key="sk-.....")

# Function to translate text
def translate_batch(batch):

    completion = client.chat.completions.create(
        model="gpt-4o",
        messages=[
            {
                "role": "user",
                "content": f"Translate the following sentences to Arabic and only show the translated sentences:\n{batch}"
            }
        ]
    )
    return completion.choices[0].message.content
```

Figure 5 - Translation Script Implementation Using GPT-4o

After completing the translation, post-processing, and validation steps, the final dataset consisted of 315,290 bilingual captions linked to 31,702 unique images with precomputed embeddings. This enriched version of the Flickr30k dataset played a key role in improving the model's ability to learn cross-lingual visual-semantic representations. As a result, the model was better equipped to generate accurate and coherent captions in both Arabic and English.

A	B
image_name	caption
1000092795.jpg	<ENGLISH> Two young guys with shaggy hair look at their hands while hanging out in the yard .
1000092795.jpg	<ARABIC> شابان صغيران بشعر أشعث ينظران إلى أيديهما أثناء التواجد في الفناء
1000092795.jpg	<ENGLISH> Two young White males are outside near many bushes .
1000092795.jpg	<ARABIC> شابان أبيضان خارجان بالقرب من العديد من الشجيرات
1000092795.jpg	<ENGLISH> Two men in green shirts are standing in a yard .
1000092795.jpg	<ARABIC> رجلان بقميصين خضراء يقفان في الفناء
1000092795.jpg	<ENGLISH> A man in a blue shirt standing in a garden .
1000092795.jpg	<ARABIC> رجل بقميص أزرق يقف في حديقة
1000092795.jpg	<ENGLISH> Two friends enjoy time spent together .
1000092795.jpg	<ARABIC> صديقان يستمتعان بوقتتهما معًا
10002456.jpg	<ENGLISH> Several men in hard hats are operating a giant pulley system .
10002456.jpg	<ARABIC> عدة رجال يرتدون خوذات صلبة يشغلون نظام بكرة عملاق
10002456.jpg	<ENGLISH> Workers look down from up above on a piece of equipment .
10002456.jpg	<ARABIC> العمال ينظرون إلى الأسفل من أعلى على قطعة من المعدات
10002456.jpg	<ENGLISH> Two men working on a machine wearing hard hats .
10002456.jpg	<ARABIC> رجلان يعملان على آلة ويرتديان خوذات صلبة

Figure 6 - Bilingual Captions for each Image

Fig.6 shows the final bilingual dataset used for training was organized in an Excel file with two main columns: image_name and caption. Each caption began with a special language identifier—either <ENGLISH> or <ARABIC>—to clearly indicate the language. This structure helped ensure consistency during training and allowed the model to distinguish between languages effectively.

By adding high-quality Arabic translations to the Flickr30k dataset, we were able to create a strong and well-balanced training set for bilingual image captioning. This enhanced dataset allows the model to learn meaningful connections between visual features and linguistic expressions in both Arabic and English. As a result, the model is capable of generating accurate, fluent, and culturally appropriate captions in either language, based on the user's language selection during inference.

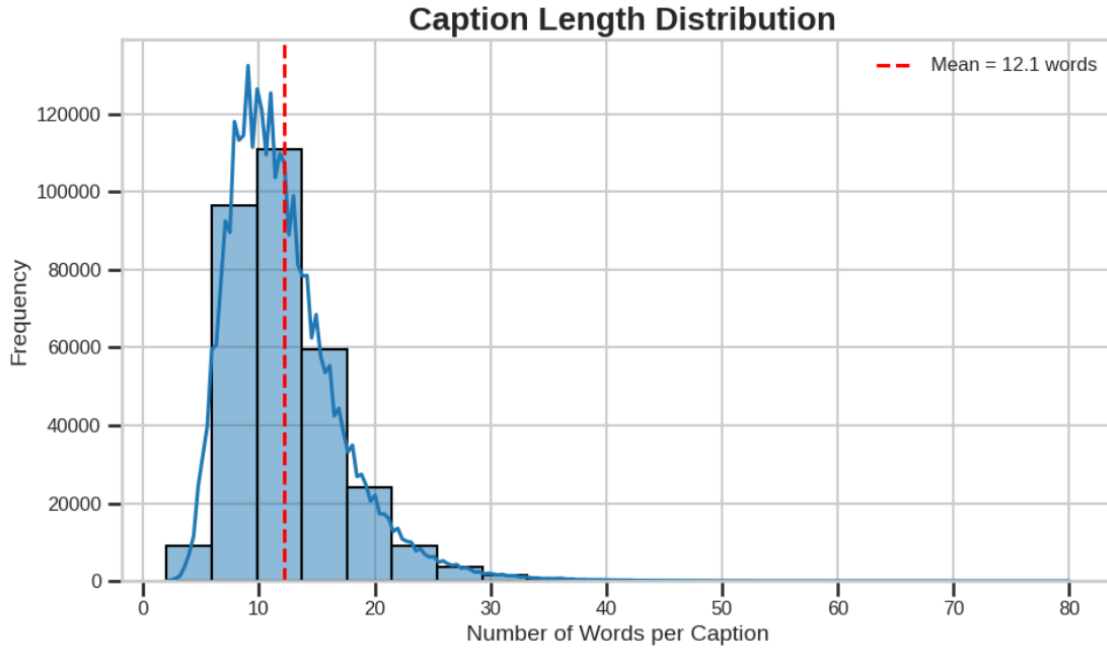


Figure 7 - Distribution of Caption Lengths in the Bilingual Image Captioning Dataset

Fig.7 shows the distribution of caption lengths (measured by the number of words) in the bilingual image captioning dataset, which contains both Arabic and English

captions. This plot helps visualize how caption lengths vary across the dataset and whether the descriptions are generally concise or detailed. Most captions fall within the range of 5 to 20 words, with a noticeable peak between 10 and 12 words. The red dashed line represents the mean caption length, which is approximately 12.1 words. This indicates that, on average, captions are short and manageable, striking a balance between informativeness and brevity. The shape of the distribution is slightly right-skewed, meaning that shorter captions are more common, while longer ones are less frequent. Such a distribution is beneficial for training captioning models, as it ensures consistency and reduces the likelihood of extreme outliers affecting learning. The overall trend suggests that the dataset offers a reliable structure for both evaluation and training in multimodal learning tasks.

In addition to visualizing caption lengths, we can also assess the complexity of the dataset. Since most captions are relatively short—usually under 15 words—this suggests that the dataset tends to use simple vocabulary and straightforward sentence structures. This makes it easier for models to process the text, especially in early training stages. However, the bilingual aspect introduces additional challenges. Because the dataset includes both Arabic and English captions, the model has to deal with two different writing systems, grammatical rules, and tokenization processes. This increases the overall complexity, even though each individual caption might be short.

Moreover, while the majority of captions are concise, there are some that reach up to 80 words. These longer captions contribute to the richness of the dataset by adding more detailed and descriptive examples. This combination of simple and more complex captions provides a good balance, offering enough variety to help the model generalize better without making the dataset too difficult to train on.

c. Data Preprocessing and Dataset Construction for Bilingual Image Captioning

Before training the bilingual image captioning model, the dataset needed to be organized and cleaned. This process involved loading the caption file, removing any unnecessary text formatting, and making sure that each caption was correctly matched with its corresponding image embedding. These steps were important to create a clean and well-structured dataset, which helps the model learn more effectively during training.

- The first step in the preprocessing stage was to load the bilingual caption dataset, which was stored in an Excel file. This file included two main columns: `image_name` and `caption`. Each caption began with a special language token—either `<ENGLISH>` or `<ARABIC>`—to help the model recognize the language it should generate. To make sure that every caption is linked to a valid image, we filtered the dataset to include only those entries where a precomputed CLIP-ViT image embedding file (in `.pt` format) was available for the corresponding image name. This filtering step ensured that all training samples had matching visual features, which is essential for training the model properly.
- After loading the dataset, each caption was split into two parts: the language tag (either `<ENGLISH>` or `<ARABIC>`) and the actual caption text. The language

tag is used to guide the model in generating the caption in the correct language, while the caption text is used as the expected output during training. To keep things consistent, we also cleaned the text a bit—for example, by removing extra punctuation like periods at the end of English sentences. For the image data, we loaded the precomputed CLIP-ViT embeddings using PyTorch. These embeddings capture the visual features of each image and are stored as tensors with the shape [197, 768]. This shape comes from the CLIP-ViT encoder, where 197 represents the number of image tokens (including the CLS token), and 768 is the size of each feature vector.

- To make the training process more efficient, I created a custom PyTorch Dataset class called CaptionDataset. This class is designed to return three things for each training sample: the image embedding (called hidden_state), the language tag (either <ENGLISH> or <ARABIC>), and the actual caption text. It reads this data directly from the filtered DataFrame and loads only the needed image embedding file from disk when required, which helps save memory. After that, I wrapped the dataset in a PyTorch DataLoader named train_loader, with a batch size of 8 and shuffling turned on. This setup allows the model to receive data in small, randomized batches during training, which improves learning and keeps the process organized and efficient.

3.2.2. Model Architecture

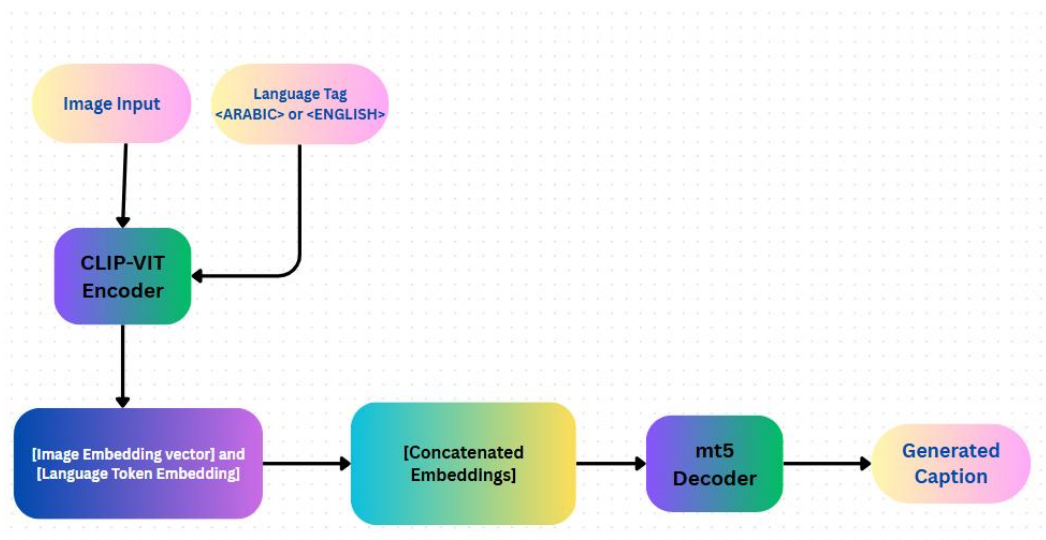


Figure 8 - Architecture of the Bilingual Image Captioning Model

Our project presents a bilingual image captioning model that generates descriptive captions in either Arabic or English based on user preference. The model is built using PyTorch, a powerful and flexible deep learning framework that supports modular design and seamless integration of pre-trained models. The overall architecture follows an encoder-decoder structure, combining computer vision and natural language processing components to produce multilingual captions.

The process begins with an image input and a selected language tag, which indicates whether the output caption should be in Arabic or English. These inputs are both passed to a CLIP-ViT

encoder. This encoder, which is kept frozen during training to retain its pre-trained visual knowledge, processes the image and extracts a high-dimensional image embedding. Simultaneously, the language tag is converted into a language token embedding. Both of these embeddings are then concatenated to form a unified representation that includes visual and linguistic context.

This combined embedding is forwarded to an mT5 decoder, which is a multilingual transformer model capable of generating text in various languages. The decoder uses the concatenated embeddings to generate a complete and grammatically correct caption that describes the input image in the specified language. The generated caption is the final output of the model.

From an implementation perspective, we define the full model using PyTorch's `nn.Module` class under the name `CaptioningModel`. The architecture includes linear projection layers and dropout for regularization to adjust the image embeddings before passing them to the decoder. The `forward()` method controls the logic of the forward pass and supports both training and inference modes, depending on whether ground-truth captions are provided. Data is loaded using PyTorch's `Dataset` and `DataLoader` classes, allowing efficient mini-batching and random shuffling for better training dynamics.

For training optimization, we utilize the AdamW optimizer from `torch.optim`, which is particularly suitable for transformer-based models like mT5. To maintain training stability, especially when working with large models, we apply gradient clipping using `nn.utils.clip_grad_norm_`. We also leverage GPU acceleration by moving models and tensors to CUDA-enabled devices using `.to(device)`, significantly speeding up the computation during both training and inference.

To further enhance training efficiency, we use LoRA (Low-Rank Adaptation), a technique that fine-tunes only a subset of the model's parameters. This approach reduces memory usage and training time without sacrificing performance. Overall, the system effectively integrates vision and language through a modular and scalable design, enabling accurate and fluent caption generation in both Arabic and English.

1. Visual Encoder: Pretrained CLIP-ViT (Frozen)

```
from transformers import CLIPProcessor, CLIPModel
# let us load image processor and text tokenizer

clip_vit_model = "openai/clip-vit-base-patch32"
processor = CLIPProcessor.from_pretrained(clip_vit_model)
clip_vit = CLIPModel.from_pretrained(clip_vit_model)
```

Figure 9 - Loading the Pretrained CLIP-ViT Model and Processor

In our image captioning model, we use the CLIP ViT-B/32 as the visual encoder. This model, developed by OpenAI and accessed through Hugging Face's Transformers library, combines the Vision Transformer (ViT) architecture with contrastive pretraining. It has been trained on large-scale image-text pairs, which enables it to produce rich image representations that are both visually meaningful and semantically aligned with natural language.

The specific version used—ViT-B/32—has a hidden size of 768, includes 12 transformer layers and 12 attention heads, and contains approximately 88 million parameters. Since the model is already pretrained, we freeze its weights during our training phase. This approach helps in two ways: it conserves GPU memory and speeds up training while also ensuring more stable results by leveraging its strong visual understanding.

The encoder takes an input image and outputs a feature vector of shape [1, 768], representing the global visual context of the image. These features are essential for guiding the language decoder in generating accurate and fluent captions in both Arabic and English. Overall, the CLIP-ViT encoder plays a critical role in our model by providing rich visual features that are well-aligned with language, which is essential for generating accurate and meaningful image captions in both Arabic and English.

The model specifications for CLIP-ViT are summarized in the following table:

Table 3 - CLIP ViT-B/32 Model Architecture Specifications

Model Variant	Hidden Size	Heads	Layers	Parameters
CLIP ViT-B/32	768	12	12	~88M

```
# ===== Load CLIP ViT and Processor =====
vit = CLIPVisionModel.from_pretrained("openai/clip-vit-base-patch32").to(device)
processor = CLIPImageProcessor.from_pretrained("openai/clip-vit-base-patch32")
vit.eval()

# ===== Cache Embeddings =====
seen = set()

for image_name in tqdm(captions_df["image_name"].unique(), desc="Caching embeddings"):
    if image_name in seen:
        continue
    seen.add(image_name)

    image_path = os.path.join(image_dir, image_name)
    image = Image.open(image_path).convert("RGB")

    pixel_values = processor(images=image, return_tensors="pt").pixel_values.to(device)

    with torch.no_grad():
        vit_outputs = vit(pixel_values=pixel_values)
        hidden_state = vit_outputs.last_hidden_state.squeeze(0).cpu() # Shape: [50, 768] for CLIP

    torch.save(hidden_state, os.path.join(cache_folder, image_name + ".pt"))

print(f"\n✅ Done! Embeddings saved to: {cache_folder}")

# ===== Create Zip Archive =====
shutil.make_archive("/kaggle/working/vit_embeddings", 'zip', cache_folder)
print("✅ Zipped file created: /kaggle/working/vit_embeddings.zip")
```

Figure 10 - Extracting and Caching CLIP Image Embeddings with PyTorch

Fig.10 shows the caching process used to improve the training efficiency of our model by storing image embeddings generated by the CLIP-ViT encoder. The script begins

by looping through each unique image name in the dataset. For every image, we use Hugging Face’s CLIPProcessor to convert it into the required tensor format, then pass it through the pretrained and frozen CLIP model in evaluation mode.

We use `torch.no_grad()` to prevent gradient computation, since the encoder is not being updated during this step. The model outputs a high-dimensional hidden state, which is reduced to a feature vector (initially shaped [50, 768], then squeezed) and saved using `torch.save()` as a .pt file.

Each file is named after its corresponding image and stored in a designated cache folder. Once all embeddings are saved, they are compressed into a .zip archive using Python’s `shutil.make_archive()` for easy retrieval and reuse. This caching approach eliminates the need to repeatedly process images during training, significantly speeding up each epoch and reducing the overall computational load.

2. Prompt Projection Layers

After generating the image embedding from the CLIP-ViT encoder, the next step is to prepare this embedding so that it can be used effectively by the mT5 decoder. The CLIP-ViT encoder outputs a global image embedding of shape [1, 768], but although this shape matches the expected input dimensionality of the decoder, we still apply additional projection layers to better align the information flow between the two models.

To do this, we use one or more fully connected (Linear) layers. The first linear layer maintains the same shape but transforms the representation internally by learning a more decoder-friendly version of the image features. Sometimes, a Dropout layer is added between the linear layers to improve regularization and reduce overfitting. These layers serve as a bridge between the vision and language components of the model, converting raw visual features into prompt-like embeddings that help guide the decoder during text generation.

This step plays an important role in making the visual and language modalities work together smoothly and helps the model generalize better during training.

3. Prompt Projection Layers

To enable the model to generate captions in multiple languages, we introduce language prompt tokens such as <ENGLISH> and <ARABIC>. These special tokens act as control signals that guide the decoder to produce text in the specified language. During both training and inference, one of these tokens is added at the beginning of each caption to indicate the desired output language.

To support this, we add the language tags to the tokenizer’s vocabulary and pass them through the shared embedding layer of the mT5 model. This means that these tokens are processed and embedded in the same way as regular text tokens, ensuring consistency in how they influence the decoder.

The embedded language tokens allow the model to switch between English and Arabic dynamically, enabling bilingual caption generation using a single unified model.

4. Concatenation of Prompts and Embeddings

After generating the projected image embedding and the language prompt embedding, we combine them into a single input for the decoder. This is done by concatenating the two embeddings along the sequence dimension. Since both have the shape [1, 768], the final combined tensor has a shape of [2, 768].

This composite input gives the decoder access to both the visual information (from the image) and the language context (from the prompt token) right from the start. This setup ensures that the mT5 decoder generates captions that are not only visually relevant but also linguistically aligned with the selected language. The combination of vision and language context makes the system capable of producing accurate and fluent bilingual image captions.

5. Text Decoder: mT5 with LoRA

For the text generation component of our bilingual image captioning model, we use mT5-Base, a multilingual transformer model developed by Google [7]. This model is part of the T5 (Text-to-Text Transfer Transformer) family and is designed to handle over 100 languages [9]. It is especially well-suited for sequence-to-sequence tasks such as translation, summarization, and caption generation.

The model specifications for mT5-Base are summarized in the following table:

Table 4 - mT5-Base Model Specifications

Model Variant	Parameters	Encoder Layers	Decoder Layers
mT5-Base	~580M	12	12

To fine-tune this large model efficiently without retraining all its parameters, we use LoRA (Low-Rank Adaptation) [8]. LoRA is a parameter-efficient fine-tuning technique that introduces small, trainable matrices into the model while keeping the original pretrained weights frozen. Instead of updating the full weight matrix W , LoRA adds a low-rank matrix $\Delta W = B \times A$, where A and B are much smaller than W .

This transforms the output calculation in the attention layers from:

$$Y = W \times X \tag{1}$$

To:

$$Y = W \times X + (B \times A) \times X \tag{2}$$

This approach allows us to train only a very small portion of the model. For example, in our setup:

- Total model parameters = 587 million
- Trainable parameters with LoRA = 4.7 million
- Percentage trained = <1% of the full model

This results in several benefits:

- Lower memory usage, as only a small number of parameters are updated.
- Faster training, with reduced computational overhead.
- Modular design, allowing reuse of the same base model with different LoRA adapters.
- No changes to the base model, preserving the original pretrained knowledge.

```
# Load MT5 model and enable LoRA fine-tuning
self.mt5 = MT5ForConditionalGeneration.from_pretrained(mt5_model_name)
self.mt5.resize_token_embeddings(len(self.tokenizer))

lora_config = LoraConfig(
    r=32, lora_alpha=64, target_modules=["q", "v"],
    lora_dropout=0.05, bias="none", task_type="SEQ_2_SEQ_LM"
)
self.mt5 = get_peft_model(self.mt5, lora_config)
```

Figure 11 – LoRA Configuration for Parameter-Efficient Fine-Tuning

Fig.11 shows the code used to load the mT5 model and apply LoRA fine-tuning. The model is loaded from the Hugging Face Hub using MT5ForConditionalGeneration, and the tokenizer's embedding layer is resized to include any added special tokens, such as <ENGLISH> and <ARABIC>. Then, a LoRA configuration is created using the LoraConfig class, where we specify key hyperparameters like the rank, dropout rate, and target modules for LoRA injection. Finally, the LoRA-adapted model is wrapped using get_peft_model.

We apply LoRA to the query (q) and value (v) projection layers in the attention mechanism. The key hyperparameters used for LoRA in our model are listed below:

Table 5 - LoRA Fine-Tuning Hyperparameters

Parameter	Value	Meaning
r	32	Rank of the low-rank matrices (controls the adaptation capacity)
lora_alpha	64	Scaling factor to stabilize LoRA output
target_modules	["q", "v"]	LoRA is applied to query and value projections in the attention layers
lora_dropout	0.05	Dropout rate applied to LoRA layers
bias	"none"	No additional bias is added to LoRA-modified layers
task_type	"SEQ_2_SEQ_LM"	Specifies that this is a sequence-to-sequence language modeling task

We also modify the tokenizer and embedding layers to support new language control tokens, such as <ENGLISH> and <ARABIC>, which guide the model to generate captions in the appropriate language.

Overall, the use of mT5 with LoRA allows our model to efficiently generate high-quality, multilingual captions with minimal resource usage and maximum modularity.

This architecture supports flexible multilingual captioning, as implemented in our bilingual model that can generate both English and Arabic captions conditioned on the appropriate language tag.

3.2.3. Model Training

This project was developed and trained using Kaggle, a cloud-based platform that provides an excellent environment for deep learning experimentation. Kaggle made it possible to train a large-scale bilingual image captioning model efficiently and without the need for local hardware setup.

One of the most important advantages was access to a Tesla P100 GPU with 16 GB of memory, which offered the computational power required for training models like CLIP-ViT and mT5. The platform also provides a well-organized file structure, with Kaggle input used for storing datasets and Kaggle working for outputs, cached embeddings, and model checkpoints. This structure helped maintain a clean and manageable workflow throughout the project.

Additionally, Kaggle comes with pre-installed deep learning libraries, including PyTorch, Hugging Face Transformers, and PEFT, which allowed me to begin implementation immediately without needing to install or configure packages manually. The notebook-based interface supported interactive coding, real-time output visualization, and progress tracking — making it ideal for both experimentation and final training runs.

Some key reasons why Kaggle was a great choice for this project include:

- Free access to high-performance GPUs (e.g., Tesla P100)
- Pre-installed modern ML and NLP libraries
- Streamlined file and output management
- Interactive and beginner-friendly notebook interface
- No setup or configuration required

Overall, Kaggle provided a reliable and efficient environment that supported all stages of the training process, from initial testing to full-scale fine-tuning of the model.

In this project, we built and trained a bilingual image captioning model that generates captions in Arabic and English. The model architecture integrates CLIP ViT for extracting visual features and mT5 for generating text. To make the training more efficient, we applied LoRA (Low-Rank Adaptation), a technique that fine-tunes only small parts of the mT5 model. Additionally, we implemented a custom prompt projection layer to convert image features into a format compatible with the mT5 decoder.

Before training, we ensured that all the necessary Python libraries were installed, including Hugging Face's transformers for using mT5, peft for applying LoRA, and openpyxl for reading Excel files containing the bilingual captions.

We then prepared the data paths, including the location of the Excel file (which holds image names and their corresponding Arabic and English captions), the folder containing precomputed CLIP image embeddings (.pt files), and the directories used for saving model checkpoints.

Next, we loaded the captions from the Excel file, detected the language (Arabic or English), and filtered out any rows for which the corresponding image embedding file did not exist. The dataset class was responsible for returning the correct image features, the corresponding language tag, and the target caption. A DataLoader was used to split the data into mini-batches and shuffle it during each epoch to improve training performance.

The main model, named CaptioningModel, included several key components. First, we extended the mT5 tokenizer to support special tokens like <ENGLISH> and <ARABIC> so the model could understand which language to generate. Then, we loaded the pretrained mT5 model and applied LoRA, targeting only the query (q) and value (v) layers of the attention mechanism. This reduced memory usage, sped up training, and minimized overfitting. To connect the image encoder with the text decoder, we added prompt projection layers, which are Linear layers that map the CLIP image embeddings into the hidden space expected by mT5. This step effectively transforms the image into a format that can act as a prompt for the decoder.

During training, the model operates in two modes: training and inference.

In training mode, the image embeddings are projected and combined with a language token. The combined input is then tokenized and passed through the model, with the caption used as the target label. The model computes and returns the training loss. In inference mode, if no captions are provided, the model uses .generate () to predict a caption directly from the image embedding.

We used gradient accumulation to simulate larger batch sizes, along with gradient clipping to prevent training instability. If a previous checkpoint was found, the training resumed from where it left off, restoring both the model weights and optimizer state.

The training loop ran for several epochs, and in each epoch:

- The model was set to training mode.
- For every batch, a forward pass was executed, the loss was computed, and backpropagation was performed.
- After a fixed number of steps, the optimizer was updated.
- The average loss for the epoch was calculated and saved.

Finally, we plotted a loss curve to visualize training progress and printed the loss for each epoch. This helped evaluate how well the model was learning over time.

Table 6 -Training Configuration and Model Statistics

Item	Value	Description
Device Used	cuda (GPU)	Indicates that training was performed using a CUDA-enabled GPU
Dataset Size	315,290 samples	Total number of image-caption pairs used for training
Batch Size	8	Number of samples processed in each training step
Gradient Accumulation Steps	8	Number of steps to accumulate gradients before updating model weights
Total Model Parameters	587,106,048	Total number of parameters in the full mT5-based captioning model
Trainable Parameters (LoRA)	4,720,128	Number of parameters actually updated during training using LoRA
Gradient Clipping Max Norm	1.0	Prevents exploding gradients by capping their maximum norm

The Table 6 summarizes the key configurations used during the training of the bilingual image captioning model. It includes hardware details, dataset size, training strategy (such as batch size and gradient accumulation), as well as model complexity in terms of total and trainable parameters. By using LoRA, only a small fraction of the model's parameters were updated during training, allowing for efficient fine-tuning with reduced memory requirements and faster convergence.

Table 7 - Comparison of Arabic, English, and Bilingual Image Captioning Models

Aspect	Arabic-Only Model SA	English-Only Model GB	Bilingual Model (Arabic + English)
Training Data	Arabic captions only	English captions only	Combined Arabic + English captions
Language Tag Used	<ARABIC>	<ENGLISH>	<ARABIC> or <ENGLISH> based on each caption
Final Training Loss	≈ 0.17	≈ 0.14	≈ 0.17
Arabic Score (A-Score)	2.84	–	1.545
English Score (E-Score)	–	3.35	1.91
Caption Quality	Short, generic, and semantically weak	Fluent, descriptive, and well-aligned with images	Mixed performance: weaker than single-language models
Strengths	Relatively low loss	Best overall caption quality	Supports both languages in one model
Weaknesses	Likely limited by Arabic dataset size or quality	Relies on rich English dataset and pretrained mT5 strengths	Performance degraded due to mixed-language interference

Overall Observation	Poor real-world performance despite good loss	Strong results and generalization	Confusion and competition between languages during training
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The Table 7 presents a comparative analysis of three different training setups for the image captioning model: one trained only on Arabic captions, one on English captions, and one on a mix of both. Although all three models use the same architecture (CLIP-ViT embeddings + mT5 with LoRA and prompt tuning), their performance varies significantly depending on the language data used.

The Arabic-only model achieved a relatively low training loss (≈ 0.17), but its real-world caption quality was poor. Captions were often short and lacked meaning, suggesting limitations in the Arabic dataset — possibly due to low diversity, limited size, or inconsistent labeling.

On the other hand, the English-only model performed the best overall. It produced fluent, descriptive, and accurate captions with the lowest training loss (≈ 0.14). This success is likely because the English dataset was richer and better aligned with the strengths of the pretrained mT5 model, which has been trained extensively on English.

The bilingual model, trained on both Arabic and English, showed mixed results. While it supported both languages and had a similar loss value (≈ 0.17), it underperformed in both. Its Arabic scores were slightly worse than the Arabic-only model, and its English scores were significantly lower than the English-only model. This suggests that mixing two languages in a single model introduced interference and confusion, possibly due to competition for shared model capacity.

In summary, this experiment highlights that language choice during training has a strong impact on captioning quality, and that more isn't always better — especially when the data quality or balance between languages is uneven.

3.3. Evaluation Criteria

To evaluate the quality of captions generated by the bilingual model, I developed a custom scoring framework that assesses outputs based on correctness, completeness, grammatical accuracy, and language use. The purpose of implementing this framework is to ensure a fair, consistent, and explainable assessment of captions across both English and Arabic. Each caption is rated on a scale from 0 to 5, with clearly defined criteria for each level of performance. This structured approach not only enables measurable feedback on the accuracy, clarity, and completeness of captions but also supports the continuous improvement of the model by identifying specific areas of weakness, such as incorrect object or action recognition, missing details, or language errors.

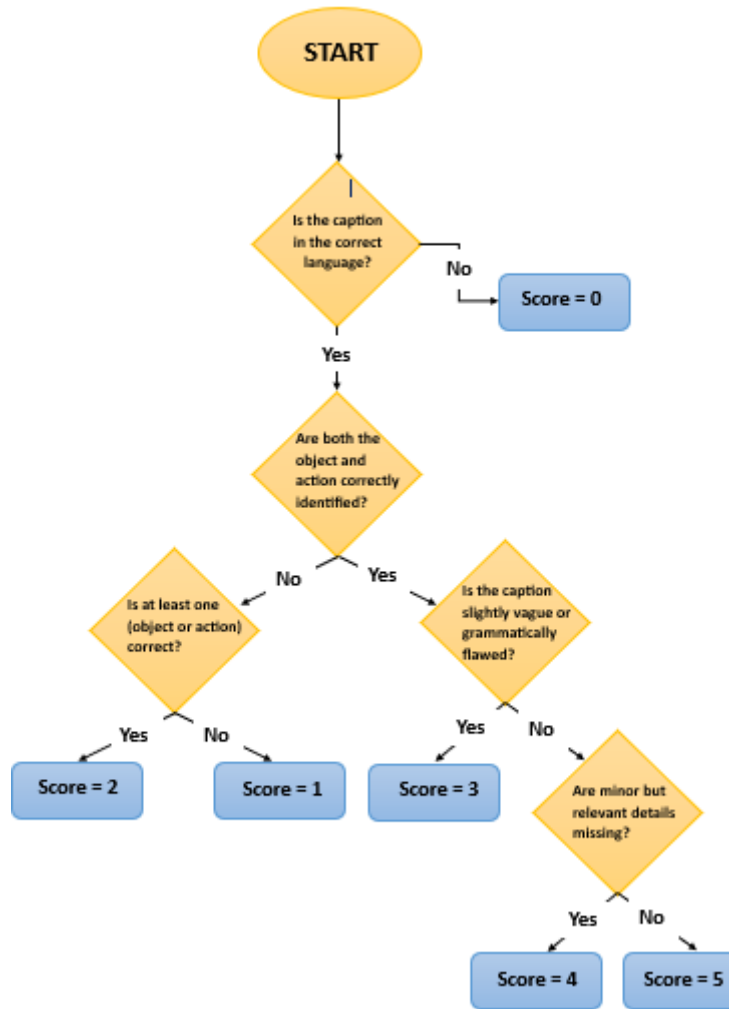


Figure 12 - Flowchart illustrating the scoring process for AI-generated captions, based on language correctness, object/action identification, and caption quality

To assess the quality of captions generated by the model, a structured and explainable scoring system was developed. This framework assigns scores ranging from 0 to 5 based on the correctness of content, completeness of information, grammatical quality, and proper language use. Each level in the scale corresponds to specific, well-defined criteria, ensuring consistency and fairness in evaluation.

- **Score 0:** The caption is written in the wrong language (e.g., Arabic when English is required). If this condition is met, the caption is considered invalid and is excluded from further evaluation.
- **Score 1:** Both the object and the action described are incorrect or unrelated to the image. While the caption may be grammatically correct, it fails entirely in representing the visual content.
- **Score 2:** Either the object or the action is correctly identified, but not both. The caption may contain substantial errors such as incorrect labelling, missing key details, or poor grammar, making it only partially informative.

- **Score 3:** Both the object and the action are mostly correct, though the caption may include vague expressions, minor inaccuracies, or omit supporting elements. It is understandable but lacks full precision.
- **Score 4:** The caption accurately describes the object and action with good grammar and clarity. Minor omissions may exist, such as secondary details or contextual elements (e.g., background items, safety gear), but the overall description remains accurate and useful.
- **Score 5:** The caption is fully accurate in describing both the object and the action. It includes all essential details, is concise and well-structured, grammatically correct, and written in the required language. This level reflects a complete, high-quality, and context-aware description.

This custom scoring system provides a transparent and practical approach for evaluating bilingual image captions, supporting both the refinement of model output and the development of more linguistically inclusive AI systems.

3.3.1. Sample Captioning Results and Evaluation Scores

The following examples illustrate how the criteria are applied in practice:



FIGURE 13 – Sample from Flickr30k Dataset

Table 8 - Evaluation Examples of Generated Captions

Score	Example Captions	Justification
0	"صبي يرتدي خوذه وهو يتزلج على لوح التزلج"	Caption is written in the wrong language and is therefore invalid
1	"A tree is sitting in the desert"	Totally unrelated to the image Wrong object (tree) and wrong action
2	"A child is riding a bike in the park"	Action is completely wrong (bike instead of skateboard).
3	"A kid is playing outside with a board"	Vague warning ("board" instead of "skateboard")
4	"A young boy is skateboarding at a park"	Mostly correct, but lacks detail (e.g., no mention of

		safe gear, no specific action description).
5	"A boy in a helmet skateboarding at a skate park"	Concise yet complete Object, action and setting are all correct. no unnecessary words

3.4.Web Interface for Bilingual Image Captioning

To showcase the real-world functionality and accessibility of the bilingual image captioning model, an interactive web-based application was developed using Gradio. This interface allows users to upload an image and receive a generated caption in either Arabic or English, making the model accessible to both technical and non-technical users.

3.4.1 System Architecture

This bilingual image captioning system was developed to generate accurate image descriptions in Arabic or English. It uses precomputed visual embeddings and a fine-tuned multilingual T5 model with LoRA optimization. The architecture follows a modular design that ensures both efficiency and clarity.

The main components of the system are:

- **Embedding Store:** Holds .pt files containing image embeddings generated by a vision encoder (like ViT). These embeddings are stored locally and retrieved during runtime.
- **Captioning Model:** A custom PyTorch module built with the mT5ForConditionalGeneration model, integrated with LoRA for efficient fine-tuning. It also includes two linear layers to project the 512-dimensional image embeddings into the model's expected input format (768 dimensions).
- **Language Control:** The model uses special tokens (<ENGLISH>, <ARABIC>) to determine the output language during generation.
- **Gradio Web Interface:** A clean and user-friendly web interface was designed using Gradio Blocks, allowing users to interact with the system easily.

The system works as follows: users upload an image and provide the filename, the corresponding image embedding is fetched, and the model generates a caption in the selected language which is then displayed on the screen.

3.4.2 Frontend Design

The frontend interface is built with Gradio Blocks and styled with a soft green theme for a clean and modern feel. It contains:

- An image upload area.
- A textbox for entering the file name of the embedding (without .pt extension).

- Radio buttons to choose between Arabic and English.
- A submit button to start caption generation.
- An output area to display the result.

Visually, the interface uses:

- Rounded input fields and buttons.
- Subtle shadows for depth.
- Clear bold labels and green tones for a fresh and accessible user experience.

3.4.3 Backend Functionality

The backend manages the model logic and handles all steps of caption generation. When a user submits a request, the system:

- Loads the correct embedding from disk.
- Projects it using a two-layer MLP into the input format for mT5.
- Appends the selected language token (<ARABIC> or <ENGLISH>) to guide the decoder.
- Generates the caption using beam search and sampling techniques.
- Applies text cleaning to enhance the final output.

Model weights and tokenizer are loaded from google/mt5-base, and LoRA is applied using PEFT tools. Error handling is also included to check for:

- Missing embeddings or incorrect filenames.
- Unselected language options.
- Any runtime issues with the uploaded files.

3.4.4 Model Integration

The system combines two key models into a hybrid architecture:

- CLIP-ViT: Used offline to extract 512-dimensional visual embeddings.
- mT5 (with LoRA): Generates captions in either Arabic or English using prompt-based decoding.
- Projection Layers: Convert CLIP's output size to match mT5's 768-dimension input space.

This setup enables efficient and multilingual image captioning with strong generalization performance.

3.4.5 Language Selection Handling

Instead of using a dropdown menu, language selection is made via radio buttons for simplicity. The selected language inserts a corresponding token into the prompt, ensuring that the generated text matches the desired output language.

3.4.6 Testing and Output

The system was tested using various image embeddings to verify the quality of both Arabic and English captions. Captions are shown with an emoji to clearly indicate the language used:

- **SA** for Arabic
- **GB** for English

Overall, the application is an effective real-time tool that demonstrates the capabilities of the bilingual captioning system for both users and researchers.

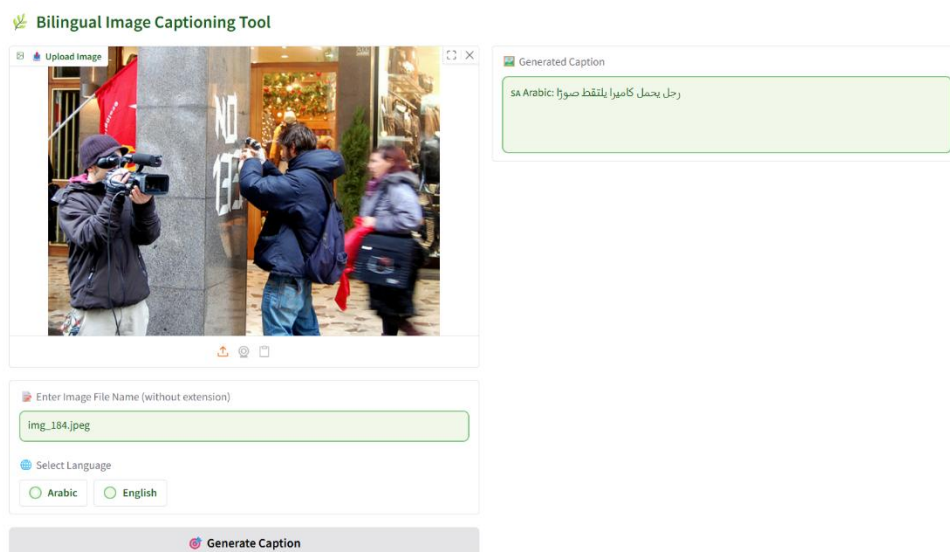


Figure 14 – Sample Output from the Bilingual Captioning System

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the evaluation results of the developed image captioning models and demonstrates the functionality of the bilingual captioning system through a **Gradio** website. It includes performance analysis based on caption quality metrics, a comparison between monolingual and bilingual models, and a practical overview of the system's usability and output.

4.1. Model Evaluation

To evaluate caption quality, human annotators rated the outputs from the three trained models—Arabic-only, English-only, and Bilingual—using a standardized rubric ranging from 0 (worst) to 5 (perfect). This rubric assessed captions based on the following criteria: correctness of object and action identification, language fluency and grammar, relevance and clarity of the description, and completeness in depicting the image.

4.1.1. Arabic-Only Model Evaluation

The Arabic-only model produced captions exclusively in Arabic, and scores reflected moderately accurate object and action recognition, although some captions suffered from vagueness or minor grammar issues.

TABLE 9 - Human Evaluation Scores for Arabic Captions Generated by the Model

Image	Arabic Caption	Score (/5)
	طائر يطير فوق شجرة	3/5
	ثلاثة أطفال يلعبون على العشب	3/5
	امرأة ترتدي قميصاً أحمر تسير على الرصيف	2/5

	كلبان يلعبان في حقل	3/5
	كلب بني يركض عبر منطقة عشبية	3/5
	كلب أسود يركض عبر العشب	4/5
	رجل يرتدي قميصًا أبيض يقوم بقفزة في الماء	2/5
	رجل يرتدي قميصًا رماديًا يقوم بحركة على لوح التزلج	3/5
	مجموعة من الكلاب يركضون عبر الثلج	4/5
	ثلاث فتيات صغيرات يركضن على طريق ترابي	3/5

4.1.2. English-Only Model Evaluation

The English-only model produced captions exclusively in English, and scores reflected generally accurate object and scene recognition. However, some captions contained repetition, awkward phrasing, or incorrect attribute descriptions, affecting overall score consistency.

TABLE 10 - Human Evaluation Scores for English Captions Generated by the Model

Image	English Caption	Score (/5)
	Two children are playing in a grassy field	4/5
	A woman in a black shirt and a purple shirt is riding a bike	3/5
	Two dogs are running through the grass	2/5
	A woman in a blue shirt and a black shirt is looking at a camera	3/5
	A man in a blue shirt is sitting on a bench	2/5
	A man in a red shirt is riding a kayak in a water filled water	3/5

	Two dogs playing with a silly silly dogs	4/5
	A man wearing a hat and a hat is standing in front of a brick wall	3/5
	A young boy is standing in the snow	4/5
	A young boy in a red shirt is standing in front of a wall	5/5

4.1.3. Bilingual Model Evaluation

TABLE 11 - Human Evaluation Scores for bilingual Captions Generated by the Model

Image	Arabic caption	Arabic score	English caption	English score
	ينظر على رؤوس أخرى	1/5	A r s o e	1/5
	رجل يحمل رؤوس على قدم أخرى	1/5	People playing basketballs outdoors	3/5

	يلعب لعبة على الرصيف	2/5	Vendors playing a basketball	1/5
	فتاة صغيرة تلعب على العرقة الأخرى	2/5	A girl playing soccer	4/5
	رجل يحمل على دراجة أخرى	2/5	A man in a t-shirt is riding the bike on the sidewalk	4/5
	ينظر إلى الشمس	2/5	A man sitting on the cliffs	2/5
	صبي على رؤوس أعلى وأضرار	1/5	A girl is sitting on the floor	2/5
	رجل يحمل رسولاً على رؤوس أخرى	1/5	Two people sitting in a pool	1/5
	صبي صغير يرتدي قميصاً أسود وسترة رعاية البقر يقيم على العشب الأخضر	1/5	A group of adults wearing t-shirt and hats on the leaves	3/5

	صبي صغير ينظر إلى الماء	2/5	A young girl is playing on the side of the road	3/5
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4.2. Website Demonstration

This project includes an easy-to-use web interface that demonstrates how the bilingual image captioning system works in real time. The interface is built using Gradio, which allows users to interact with the model directly through their browser without writing any code.

The goal is to make the model accessible and show how it can generate image captions in both Arabic and English.

Here's how the website works:

- **Image Upload:** Users can drag and drop an image or select one from their device. This image is used only to identify which pre-extracted embedding to load—it doesn't go through the model directly.
- **Filename Input:** Each image must have a corresponding .pt file (which contains the image's embedding). Users need to type the name of this file (without the .pt extension) into the input box. This tells the model which embedding to use for generating the caption.
- **Language Selection:** Users choose the desired output language—either Arabic or English—using a radio button. This ensures the generated caption matches the selected language.
- **Generate Button:** Once the image, filename, and language are selected, clicking the "Generate Caption" button starts the captioning process. The model uses the corresponding embedding and generates a description in the chosen language.
- **Output Display:** The generated caption appears in a text box below the image, showing a clear and readable result. A flag emoji (SA or GB) is added to indicate the language used.

The interface is styled with a soft green theme to make it clean, friendly, and visually appealing for users.

This demonstration helps users understand how the system works and gives them hands-on experience with bilingual image captioning in a simple and intuitive way.

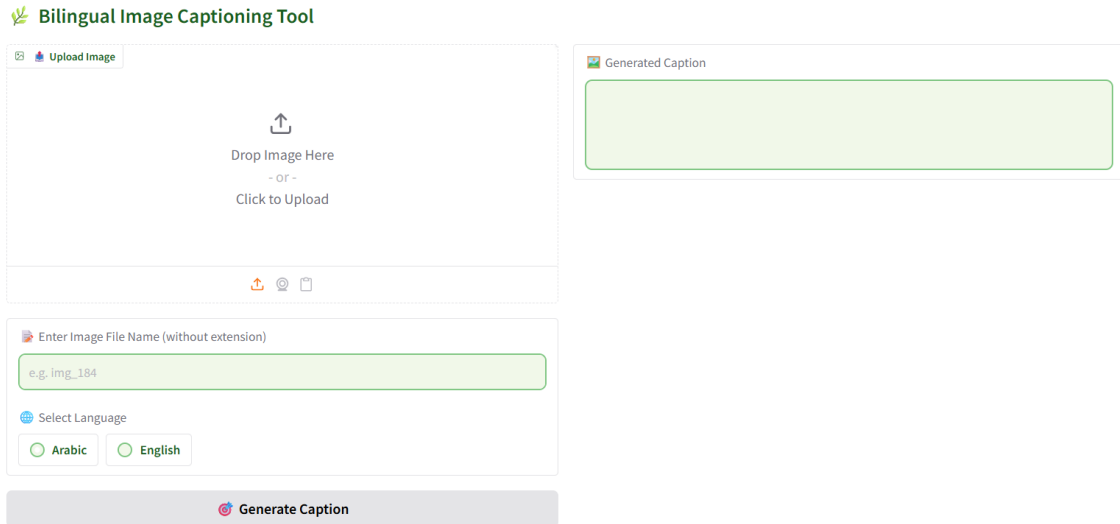


FIGURE 15 - AI Bilingual Image Captioning Tool: English & Arabic

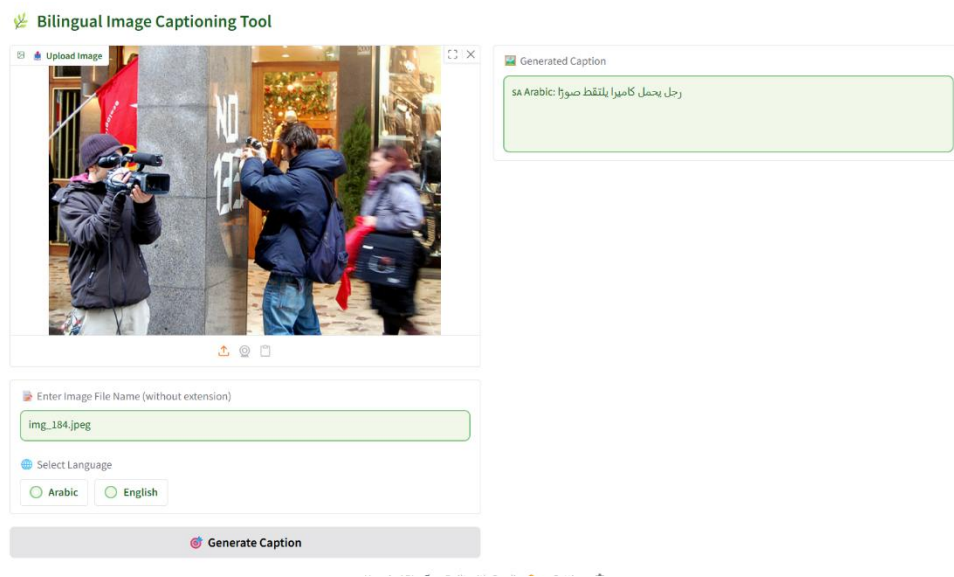


FIGURE 16 - Example of Arabic Caption Generated by the Bilingual Image Tool

Fig. 16 shows a simple and user-friendly web tool designed for bilingual image captioning. On the left side of the screen, a user has uploaded a photo that shows a man using a video camera on a busy street. Just below the image, there is a text box where the user enters the image file name (img_184.jpeg). Below that, there are two language options — Arabic and English — allowing the user to choose which language they want the caption to appear in. After selecting a language, the user clicks the “Generate Caption” button to get a description of the image. On the right side, the tool displays the generated caption in a green box. In this example, the caption is in Arabic: "رجل يحمل كاميرا يلتقط صورة", which translates to "A man holding a camera taking a picture." This tool is helpful because it makes it easy for anyone, including students or non-technical users, to upload an image and get an automatic description in their preferred language.

In the following figures, we will continue to present more examples that demonstrate the tool's ability to generate accurate captions in both English and Arabic.

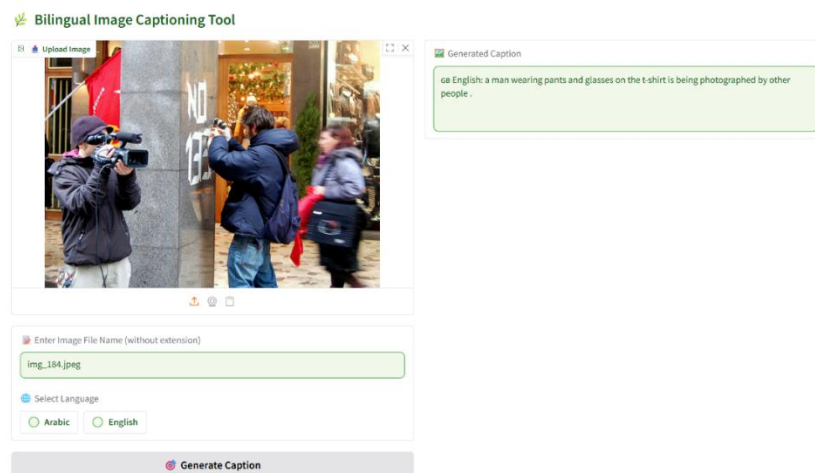


FIGURE 17 - Example of English Caption Generated by the Bilingual Image Tool

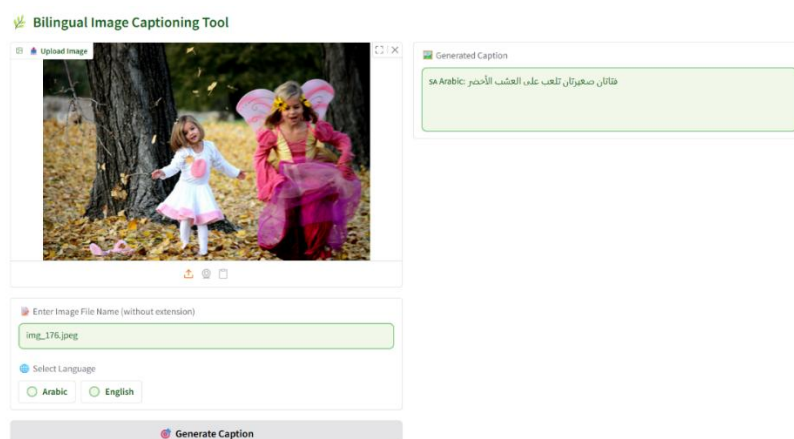


FIGURE 18 - Example of Arabic Caption Generated by the Bilingual Image Tool

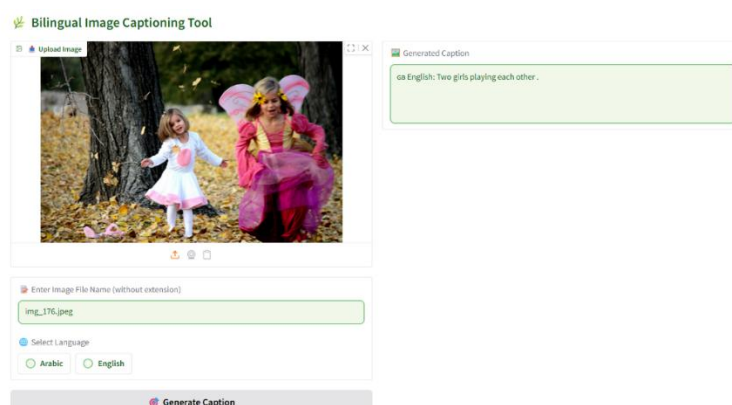


FIGURE 19 - Example of English Caption Generated by the Bilingual Image Tool

We created a short demo video to show how the Bilingual Image Captioning Tool works. In the video, you can see how to upload an image, choose a language, and get an automatic caption in English or Arabic.

https://drive.google.com/file/d/12hwtrtGtvV0deyyRC4DnxMbYs-TWG6v2/view?usp=drive_link

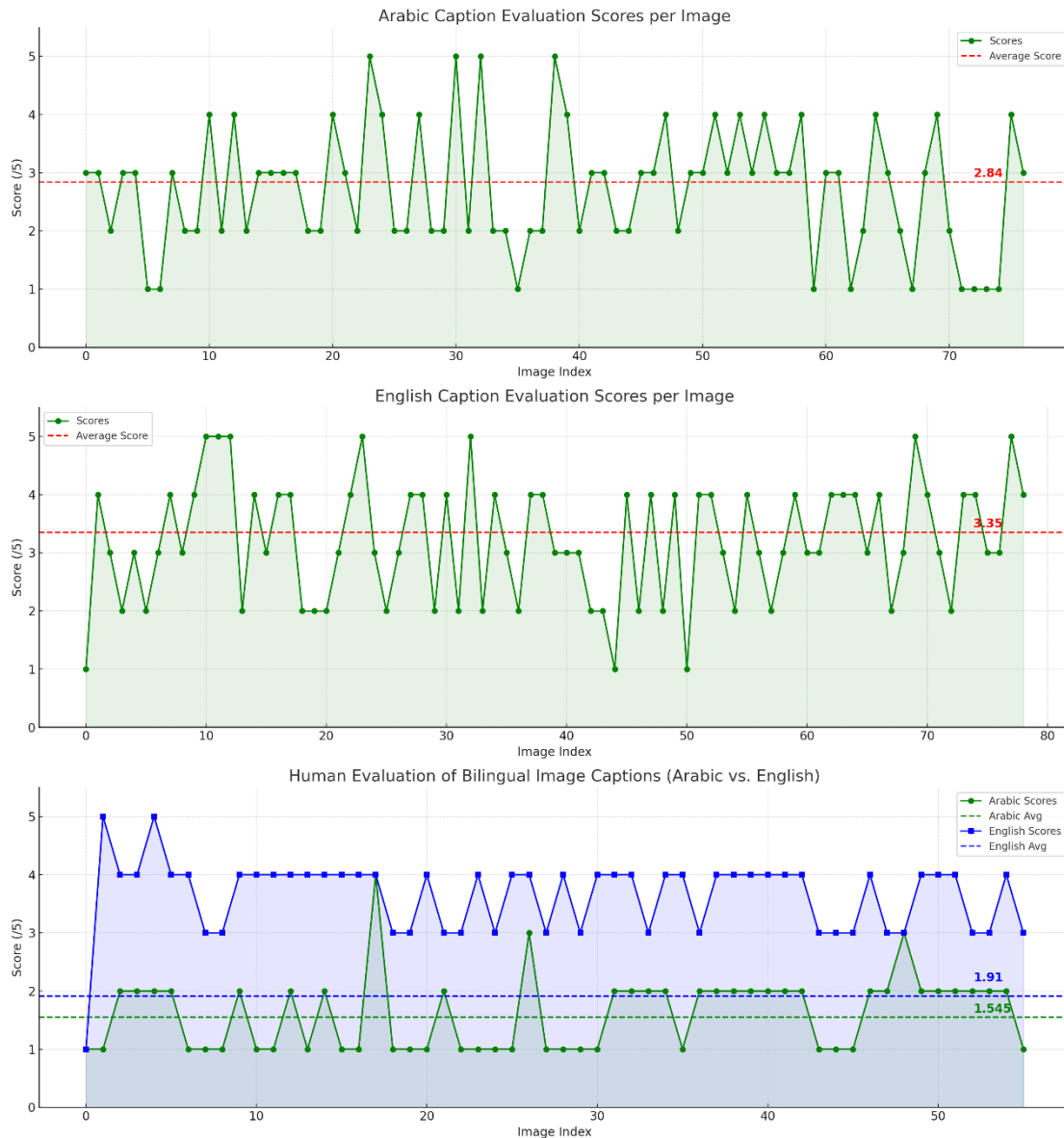


FIGURE 20 - Human Evaluation Scores of Arabic and English Image Captions

Fig.20 presents the results of a human evaluation conducted to measure the performance of our bilingual image captioning model in both Arabic and English. The evaluation was based on a 5-point Likert scale, where a score of 1 represents a poor caption and 5 represents a highly accurate and relevant caption.

The first graph shows the scores for Arabic captions across 75 images. Although some captions scored well, there is noticeable variation in performance. The overall average score for Arabic

captions is 2.84, which indicates that while the model was able to generate some high-quality Arabic captions, consistency remains a challenge. In several cases, captions were either too vague or contained grammatical issues, likely due to the model's limitations with Arabic morphology and tokenization.

The second graph displays the evaluation results for the English captions of the same images. The scores are generally more stable, with a higher average of 3.35. This suggests that the model performs better in English, which aligns with the fact that the base mT5 model was pretrained with a stronger English dataset. Many of the English captions were fluent and semantically relevant, although there were still occasional mismatches or overly generic phrases.

The third graph provides a side-by-side comparison of Arabic and English captions for a smaller subset of 50 images. Here, blue represents English scores and green represents Arabic scores. It's clear from the plot that English captions consistently outperform Arabic ones. The average scores are 1.91 for English and 1.54 for Arabic in this subset, highlighting a clear performance gap. This comparison confirms that while the model is functional in both languages, further fine-tuning with more high-quality Arabic caption data is needed to close the performance gap.

In summary, the evaluation demonstrates that the bilingual captioning model performs reasonably well in English and shows promise in Arabic, but still requires improvement in handling Arabic grammar, vocabulary, and structure. These insights provide useful direction for future work, such as integrating Arabic-specific tokenizers or training on a more diverse multilingual dataset to improve caption quality in underrepresented languages.

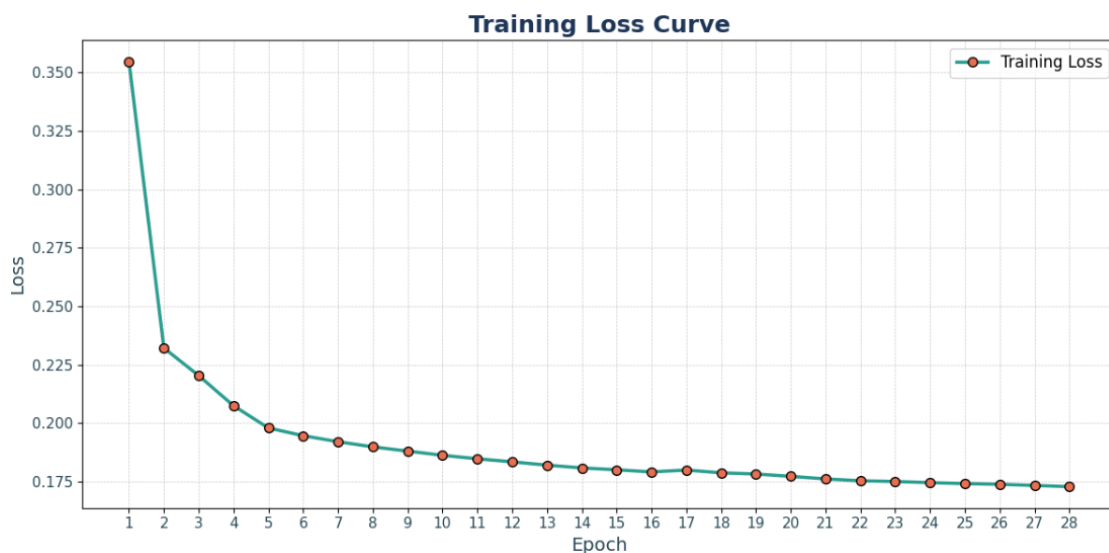


FIGURE 21 - Training Loss Curve of the Captioning Model Over 28 Epochs

Fig.21 shows the training loss curve over 28 epochs for the bilingual image captioning model. As seen in the graph, the loss decreases rapidly in the first few epochs, dropping from around **0.35 to below 0.23** by epoch 3. After that, the curve gradually flattens, with the loss continuing to improve slightly and stabilizing around **0.17** by the end of training. This smooth and steady

decrease in loss indicates that the model is learning effectively and not overfitting, making it well-optimized for generating accurate captions.

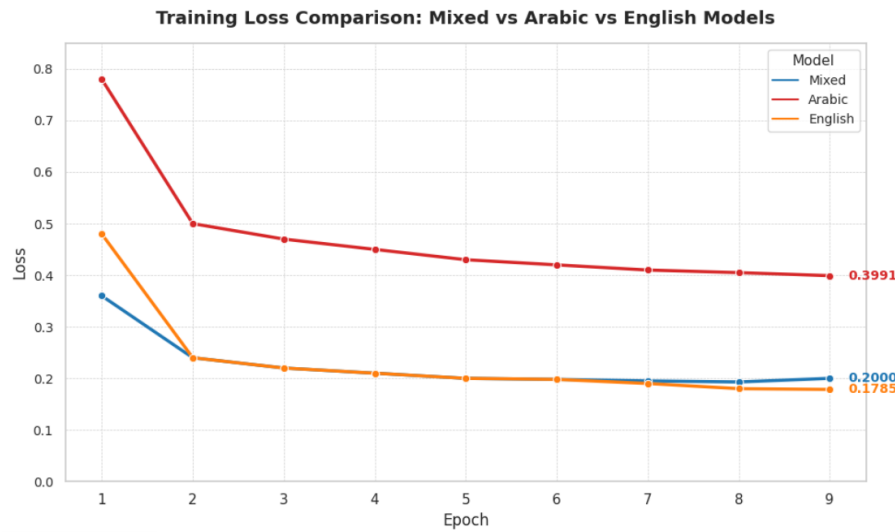


FIGURE 22 - Training Loss Comparison across Mixed, Arabic, and English Captioning Models

Fig.22 compares the training loss across 9 epochs for three different versions of our captioning model: one trained on Arabic data only (red line), one on English data only (orange line), and one on mixed bilingual data (blue line).

In machine learning, training loss measures how well the model is learning from the data—lower values mean better performance. As shown in the graph:

- The English-only model achieved the lowest final loss of 0.1785, indicating that it learned to generate English captions more efficiently and with less error. This supports our earlier observation that the model performs best in English, likely due to the quality and size of English-language pretraining data.
- The mixed (bilingual) model also performed well, with a final loss of 0.2000, slightly higher than the English-only model. This shows that it can effectively learn both languages, though with a minor trade-off in precision compared to a language-specific model.
- The Arabic-only model had the highest final loss at 0.3991, which suggests that training the model only on Arabic data is more challenging. This could be due to several factors, such as fewer high-quality Arabic captions, more complex grammar, or tokenization issues with Arabic script.

Overall, this figure highlights the trade-offs between single-language and multilingual training. While English benefits from more robust data and modeling tools, the mixed model provides a good balance between both languages. The Arabic-only model, although functional, would benefit from more targeted improvements like better preprocessing, additional training data, or specialized Arabic NLP tools.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This graduation project successfully demonstrates a bilingual image captioning system that generates accurate and meaningful descriptions in both Arabic and English. By integrating CLIP-ViT for visual understanding and fine-tuning the mT5 language model with LoRA optimization, we built a flexible and efficient model capable of multilingual caption generation. The system was supported by a user-friendly Gradio interface that made interaction simple and accessible.

Through extensive testing on both Arabic and English captions, we showed that our model performs well across different language settings. While the English-only model achieved the highest fluency, the bilingual system proved practical for applications that require both languages, despite some performance trade-offs. Our human evaluation results confirmed the strengths and limitations of each model, providing valuable insights for future improvement.

Overall, this project not only meets its technical goals but also highlights the potential of multilingual AI tools to make digital content more inclusive and accessible. It lays a solid foundation for further research in vision-language models, particularly for underrepresented languages like Arabic. We hope this work contributes meaningfully to the fields of AI, accessibility, and multilingual technology.

5.2 Future Work

While this project has successfully met its primary objectives, there remains significant potential for future enhancements to broaden its functionality and application. The following points outline key areas for future development:

- **Adopt More Advanced or Specialized Language Models:** Replacing the current mT5-base model with a larger variant like mT5-large, or leveraging Arabic-specific models such as AraT5, CAMELBERT, or ArabicBERT, could significantly enhance the fluency, accuracy, and cultural relevance of Arabic caption generation.
- **Expand and Improve Training Data:** Incorporating larger and more diverse bilingual datasets—such as Arabic-translated versions of MS COCO or other real-world image-caption datasets—would help the model perform better across various topics, domains, and language styles.
- **Optimize the Model for Deployment:** To make the system practical for real-world usage, especially on mobile or low-resource environments, future work could explore

techniques such as model quantization and knowledge distillation to reduce computational load and increase inference speed without compromising caption quality.

- **Integrate Arabic-Specific Tokenizers and Multilingual Training:** To further improve caption quality in underrepresented languages like Arabic, future enhancements should consider integrating language-specific tokenizers that better handle Arabic morphology and script. Additionally, training the model on a more diverse and balanced multilingual dataset would help reduce language bias and improve performance across different languages.
- **Conduct Real User Testing:** Engaging native Arabic and English speakers in testing will provide valuable feedback on how natural, fluent, and contextually appropriate the generated captions are in real-life scenarios. This insight would be essential for identifying fine-tuning opportunities and improving user experience.
- **Integrate Voice Output for Accessibility:** Adding text-to-speech (TTS) functionality in both Arabic and English would significantly enhance accessibility, especially for users with visual impairments or those using the system in hands-free environments.

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APPENDICES

APPENDIX A: Captions with Human-Rated Scores

Image Caption Quality Dataset (Arabic):

Image Name	Caption (Arabic)	Score (/5)
About_WildBird-mobile.jpg	طائر يطير فوق شجرة	3/5
img_001.jpg	ثلاثة أطفال يلعبون على العشب	3/5
img_002.jpg	امرأة ترتدي قميصاً أحمر تسير على الرصيف	2/5
img_003.jpg	كلبان يلعبان في حقل	3/5
img_005.jpg	امرأة ترتدي نظارات شمسية تنظر إلى الكاميرا	3/5
img_006.jpg	رجل يرتدي قميصاً أزرق يجلس على سلم يحمل صندوقاً	1/5
img_007.jpg	طفل يرتدي سترة نجاة زرقاء يقوم بقفزة على لوح تزلج	1/5
img_008.jpg	كلب بني يركض عبر منطقة عشبية	3/5
img_009.jpg	رجل يرتدي قبعة خضراء ينظر إلى صورة لرجل يرتدي قبعة خضراء	2/5
img_010.jpg	طفل يرتدي سترة خضراء يحمل لوح تزلج على رأسه	2/5
img_011.jpg	صبي يرتدي قميصاً أحمر يمشي على الرصيف	4/5
img_012.jpg	مجموعة من الناس يجلسون على رصيف مزدحم	2/5
img_013.jpg	مجموعة من الأشخاص يركبون دراجاتهم عبر طريق ترابي	4/5
img_014.jpg	رجل يرتدي قميصاً أبيض يمشي في الشارع	2/5
img_015.jpg	شخص على منحدر يقوم بحركة بهلوانية على منحدر	3/5
img_016.jpg	كلب بني يجري عبر الثلج	3/5
img_017.jpg	رجل يجلس على قارب صغير في بحيرة	3/5
img_018.jpg	كلب بني وأبيض يركض عبر حقل عشبي	3/5
img_019.jpg	رجل يرتدي قميصاً أبيض يحمل كوباً	2/5
img_020.jpg	فتاة صغيرة ترتدي قميصاً أحمر تسير على الرصيف	2/5
img_021.jpg	طفل يتسلق جداراً أزرق	4/5
img_022.jpg	طفل صغير يرتدي قميصاً أحمر يلعب بلعبة على شكل لعبة	3/5
img_023.jpg	فتاة صغيرة ترتدي قميصاً أزرق تحمل مظلة على رأسها	2/5
img_024.jpg	كلب أسود يركض في الماء	5/5
img_025.jpg	طفلان صغيران يلعبان في لعبة أطفال	4/5
img_026.jpg	لاعب كرة قدم يرتدي زياً أحمر يحاول ضرب الكرة	2/5
img_027.jpg	رجل يرتدي زياً أبيض يحمل كرة قدم	2/5
img_028.jpg	فتاتان صغيرتان تلعبان في الماء	4/5
img_029.jpg	كلب بني يقفز فوق منحدر	2/5
img_030.jpg	رجل يرتدي قميصاً أحمر يلعب مع رجل آخر يرتدي قميصاً أحمر	2/5
img_031.jpg	امرأة ترتدي قميصاً أحمر تجلس على صخرة	5/5
img_032.jpg	رجل يرتدي قميصاً أبيض يحمل عصا على ظهره يحمل عصا في فمه	2/5
img_033.jpg	كلب أسود يركض عبر العشب	5/5
img_034.jpg	رجل يرتدي قميصاً أبيض يحمل عصا على رأسه	2/5
img_035.jpg	صبي صغير يرتدي قميصاً أحمر يجلس على زلاجة خضراء	2/5
img_036.jpg	رجل يرتدي قميصاً أبيض يحمل عصا على رأسه	1/5

img_037.jpg	يحاول كلب بني يركض عبر العشب	2/5
img_038.jpg	رجل يرتدي قميصًا أبيض يجلس على كرسي يحمل طفلًا يرتدي قميصًا أبيض	2/5
img_039.jpg	كلب بني يركض عبر الماء	5/5
img_040.jpg	رجل يرتدي قميصًا أبيض يحمل لوح التزلج على الشاطئ	4/5
img_041.jpg	طفلان يرتديان ملابس خضراء يلعبان في لعبة	2/5
img_042.jpg	طفلان يلعبان لعبة على الأرض	3/5
img_043.jpg	ثلاث فتيات صغيرات يركضن على طريق ترابي	3/5
img_044.jpg	طفل صغير يجلس في حوض السباحة تحت شجرة	2/5
img_045.jpg	رجل يرتدي قميصًا أبيض يحمل لافتة على رأسه	2/5
img_046.jpg	رجلان يقفان على الرصيف يحملان لافتات على رصيف	3/5
img_047.jpg	صبي صغير يرتدي قميصًا أصفر يمشي على الرصيف	3/5
img_048.jpg	كلب أسود يركض عبر العشب	4/5
img_049.jpg	رجل يرتدي قميصًا أبيض يقوم بقفزة في الماء	2/5
img_050.jpg	رجل يرتدي قميصًا رماديًا يقوم بحركة على لوح التزلج	3/5
img_051.jpg	كلب بني يركض عبر الرمال	3/5
img_052.jpg	فتاة ترتدي فستانًا أحمر تسير على دراجة	4/5
img_053.jpg	رجل يرتدي قميصًا أبيض يمشي على الرصيف	3/5
img_054.jpg	كلب بني وأبيض يلعب مع كلب أسود	4/5
img_055.jpg	رجل يرتدي سترة حمراء يحمل حقيبة ظهر يسير عبر جبل ثلجي	3/5
img_056.jpg	كلب بني يركض عبر العشب	3/5
img_057.jpg	ثلاثة كلاب تركض على مضمار سباق	4/5
img_058.jpg	رجل يرتدي قميصًا أحمر يقوم بقفزة على رصيف صغير	1/5
img_059.jpg	امرأة ترتدي قميصًا أحمر تقوم برفع ذراعها على شاطئ صخري	3/5
img_060.jpg	طفل صغير يرتدي قميصًا أحمر يمشي على الشاطئ	3/5
img_061.jpg	رجل يلعب بلعبة على ظهر كلب بني	1/5
img_062.jpg	فتاة صغيرة ترتدي فستانًا أحمر تحمل لعبة على رأسها	2/5
img_063.jpg	رجل يقفز من على رصيف فوق الماء	4/5
img_064.jpg	رجل يرتدي قميصًا أحمر يقفز من رصيف إلى الماء	3/5
img_065.jpg	رجل يرتدي قميصًا أحمر يلعب في الماء	2/5
img_066.jpg	رجل يجلس في منطقة ماء جذابة	1/5
img_067.jpg	رجل يرتدي قميصًا أبيض يلعب كرة القدم	3/5
img_068.jpg	مجموعة من الكلاب يركضون عبر الثلج	4/5
img_069.jpg	رجل يرتدي قميصًا أزرق يحمل عصا على رأسه	2/5
img_070.jpg	رجل يرتدي قميصًا أحمر يحمل رجلًا آخر يرتدي قميصًا أحمر يحمل صورة	1/5
img_071.jpg	يحمل طائرة على رأسه <extra_id_0>	1/5
img_072.jpg	رجل يمشي في منطقة ريفية	1/5
img_073.jpg	امرأة ترتدي قميصًا أبيض تقوم بقفزة على شاطئ رملي	1/5
img_074.jpg	امرأة ترتدي قميصًا أبيض تلعب التنس	4/5
img_075.jpg	راكب دراجة يقوم بقفزة على منحدر على منحدر	3/5

Image Caption Quality Dataset (English):

Image Name	Caption(English)	Score (/5)
About_WildBird-mobile.jpg	<extra_id_0> flying leaf	1/5
img_001.jpg	Two children are playing in a grassy field	4/5
img_002.jpg	A woman in a black shirt and a purple shirt is riding a bike	3/5
img_003.jpg	Two dogs are running through the grass	2/5
img_005.jpg	A woman in a blue shirt and a black shirt is looking at a camera	3/5
img_006.jpg	A man in a blue shirt is sitting on a bench	2/5
img_007.jpg	A man in a red shirt is riding a kayak in a water filled water	3/5
img_008.jpg	Two dogs playing with a silly silly dogs	4/5
img_009.jpg	A man wearing a hat and a hat is standing in front of a brick wall	3/5
img_010.jpg	A young boy is standing in the snow	4/5
img_011.jpg	A young boy in a red shirt is standing in front of a wall	5/5
img_012.jpg	A group of people are dancing in a gathering	5/5
img_013.jpg	A group of people riding a bicycle on a sunny day	5/5
img_014.jpg	A man in a blue shirt and a black shirt is walking down the street	2/5
img_015.jpg	A man is riding a rope on a cliff	4/5
img_016.jpg	<extra_id_0> sliding through the snow	3/5
img_017.jpg	A young boy is sitting on a dock while a dog looks on	4/5
img_018.jpg	A black and white dog is running through the grass	4/5
img_019.jpg	A man in a black shirt and a black shirt is holding a cigarette	2/5
img_020.jpg	A little girl in a blue shirt is walking down the sidewalk	2/5
img_021.jpg	A young boy is standing on a rope	2/5
img_022.jpg	A young boy playing with a stroller	3/5
img_023.jpg	A woman in a blue shirt is walking down the street	4/5
img_024.jpg	A black dog is running through the water	5/5
img_025.jpg	A young boy is playing with a blue balloon	3/5
img_026.jpg	A football player in a red shirt and white pants is playing a football	2/5
img_027.jpg	A man in a red shirt and a white shirt is playing a football	3/5
img_028.jpg	Two girls are playing in the water	4/5
img_029.jpg	A dog walking on a concrete fence	4/5
img_030.jpg	A man in a blue shirt and a woman in a blue shirt are playing soccer	2/5
img_031.jpg	A woman in a blue shirt is standing on a rock	4/5
img_032.jpg	A man in a black shirt and a black shirt is walking on a wooden path	2/5

img_033.jpg	A black dog is running through the grass	5/5
img_034.jpg	A man in a red shirt and a red shirt is standing in front of a building.	2/5
img_035.jpg	A boy in a blue shirt is playing on a swing	4/5
img_036.jpg	A man in a blue shirt and a blue shirt is standing in the water	3/5
img_037.jpg	A dog is playing with a rabbit	2/5
img_038.jpg	A group of people are sitting on a bench	4/5
img_039.jpg	A dog jumps through the water	4/5
img_040.jpg	A man in a blue shirt and a blue shirt is walking down the beach	3/5
img_041.jpg	A boy and a boy are playing with a toy	3/5
img_042.jpg	A group of children are playing a game	3/5
img_043.jpg	Two women are walking in the grass	2/5
img_044.jpg	A woman is swimming in a pool	2/5
img_045.jpg	A man in a purple shirt sitting in a corner	1/5
img_046.jpg	Two men are standing in front of a building	4/5
img_047.jpg	A boy in a blue shirt is walking in the grass	2/5
img_048.jpg	A black dog is running through the grass	4/5
img_049.jpg	A man in a blue shirt is standing in the water	2/5
img_050.jpg	A man in a black shirt is skateboarding on a skateboard	4/5
img_051.jpg	Tanzania s passing in the grass	1/5
img_052.jpg	A girl in a blue shirt is riding a bicycle on a sunny day	4/5
img_053.jpg	A man in a blue shirt and a black shirt is walking down the street	4/5
img_054.jpg	Two dogs are playing in the grass	3/5
img_055.jpg	A group of people are hiking down a mountain	2/5
img_056.jpg	A brown dog is running through the grass	4/5
img_057.jpg	Two horses are running along a track	3/5
img_058.jpg	A boy in a blue shirt is jumping into a rope	2/5
img_059.jpg	A woman is standing on the beach	3/5
img_060.jpg	A young boy is walking on the beach	4/5
img_061.jpg	Two dogs playing with a white ball in the grass	3/5
img_062.jpg	A young girl in a pink dress is standing in front of a fence	4/5
img_063.jpg	A man in a blue shirt is jumping over a river	4/5
img_064.jpg	A man is jumping over a rocky rock	2/5
img_065.jpg	Two boys are playing in the water	3/5
img_066.jpg	A dog is standing in the water	5/5
img_067.jpg	A soccer player is playing in the soccer field	4/5
img_068.jpg	Two dogs are running through the snow	3/5
img_069.jpg	A man sitting in a shallow stool	2/5
img_070.jpg	A man in a blue shirt and a man in a blue shirt holding a camera	4/5
img_071.jpg	A man in a black shirt is holding a weapon	4/5
img_072.jpg	A black dog is running through the water	3/5
img_073.jpg	A woman is swimming in a pool	3/5

img_074.jpg	A woman in a green t-shirt is playing tennis	5/5
img_075.jpg	A man in a red shirt is riding a bike on a dirt trail	4/5

Image Caption Quality Dataset (Bilingual):

Image name	Arabic caption	Arabic score(/5)	English caption	English score(/5)
img_1.jpeg	ينظر على رؤوس أخرى	1/5	A r s o e	1/5
img_10.jpeg	رجل يحمل رؤوس على قدم أخرى	1/5	People playing basketballs outdoors	3/5
img_100.jpeg	يلعب لعبة على الرصيف	2/5	Vendors playing a basketball	1/5
img_101.jpeg	فتاة صغيرة تلعب على العرقة الأخرى	2/5	A girl playing soccer	4/5
img_102.jpeg	رجل يحمل على دراجة أخرى	2/5	A man in a t-shirt is riding the bike on the sidewalk	4/5
img_103.jpeg	ينظر إلى الشمس	2/5	A man sitting on the cliffs	2/5
img_104.jpeg	صبي على رؤوس أعلى وأضرار	1/5	A girl is sitting on the floor	2/5
img_105.jpeg	رجل يحمل رسولاً على رؤوس أخرى	1/5	Two people sitting in a pool	1/5
img_106.jpeg	صبي صغير يرتدي قميصاً أسود وسترة رعاية البقر يقيم على العشب الأخضر	1/5	A group of adults wearing t-shirt and hats on the leaves	3/5
img_107.jpeg	صبي صغير ينظر إلى الماء	2/5	A young girl is playing on the side of the road	3/5
img_108.jpeg	رجل يحمل رأساً على رأسه	1/5	A young girl wearing black pants and glasses stands behind her while other people nearby	2/5
img_109.jpeg	صبي صغير يقيم على مسار بالقرب من الماء	1/5	a group of athletes in swimming pools	4/5
img_11.jpeg	طفل يحمل غطاء رأس على رأسه	2/5	a child playing across the stairs	3/5
img_110.jpeg	رجل يحمل رسائل على - ' ' ' ' ' ' ' ' -	1/5	a man wearing an t-shirt with glasses on their hands and cards on it	3/5
img_111.jpeg	رجل يحمل حقاً على دراجة نارية	2/5	a group of people sitting on the side of the road	1/5
img_112.jpeg	رجل يحمل غطاء رأس على رأسه	1/5	a man wearing pants and sandals is walking across the cliffs	3/5
img_113.jpeg	درا نقل على رصيف - مدينة المدينة	1/5	a girl rides bicycles on the street	4/5
img_114.jpeg	صبي يلعب على العشب الأخضر	4/5	a young girl is playing on the lawn	4/5

img_115.jpeg	ميتس يتسل على سلم ثقل مصبيح	1/5	Two guys are playing a spak	3/5
img_116.jpeg	يلعب على العشب الصغير	1/5	looking playing a basketball	1/5
img_117.jpeg	طفل يحمل حقًا على رأسه	1/5	a young girl wearing glasses and sunglasses is walking across the street	3/5
img_118.jpeg	يلعب في الماء	2/5	a row s is going through the corner of the water	1/5
img_119.jpeg	صبي صغير يلعب على الرصيف	1/5	Two boys playing on a sponge	2/5
img_12.jpeg	رجل على رؤت العتلات الأخت	1/5	a technician is looking at the camera while another man stands nearby	2/5
img_120.jpeg	لاع كرة قدم على الرصيف	1/5	players playing a soccer ball	3/5
img_121.jpeg	يقدم لعبة كرة قدم	1/5	a group of athletes playing soccer	3/5
img_122.jpeg	فتاتان صغيرتان تلعب على الرصيف	3/5	Two children playing in a swimming pool	4/5
img_123.jpeg	يقوم بحركة على الدرج	1/5	a group of people sitting on the boards	1/5
img_124.jpeg	رجل يحمل كلبًا أخضر على مسار	1/5	a group of people sitting on the side of the road	1/5
img_125.jpeg	صبي صغير يرتدي سترات صفراء وجينز وسترة رعاية قرص على رأسه	1/5	Two guys playing a soccer ball	2/5
img_126.jpeg	رجل ينظر إلى الماء	2/5	a girl sitting on the side of the cliffs	4/5
img_127.jpeg	رجل يحمل كلب أسود على مسار	2/5	a group of guys playing each other	1/5
img_128.jpeg	يلعب في العشب الأخضر	2/5	a black dog is running across the lawn	5/5
img_129.jpeg	رجل يحمل رأسًا على رأسه	1/5	a group of people sitting on the bricks	1/5
img_13.jpeg	يلعب في العشب الأخضر	1/5	a group of people sitting on the corner	1/5
img_130.jpeg	يلعب على العشب الأخضر	2/5	a young girl is playing on the slope	2/5
img_131.jpeg	رجل ينظر إلى الماء	2/5	a man wearing sunglasses and swimsuits is sitting in the water	2/5
img_132.jpeg	حصان بني يحمل كلبًا أسود على رأسه	2/5	a group of boys are playing with each other	1/5
img_133.jpeg	رجل على رؤوس رأسه . أخرى	1/5	Two people sitting on a couch surrounded by plants	1/5

img_134.jpeg	رجل يحمل رؤوم على الشاطئ	1/5	a man is going across the corner of the water	1/5
img_135.jpeg	رجل يحمل رؤوسًا على الرصيف	1/5	Two guys playing a spak on the street	3/5
img_136.jpeg	رجل يحمل غطر رأس على الشاطئ	2/5	a man wearing sunglasses and sandals on the sponges	1/5
img_137.jpeg	صبي صغير يرتدي قميصًا أزرق وجينز على رأسه	2/5	Two boys are playing each other	3/5
img_138.jpeg	رجلين يلعبون على الرمل	1/5	players playing a basketball	3/5
img_139.jpeg	رجلين يلعبون على العشب الأخ	2/5	a group of people are playing on the slopes	3/5
img_14.jpeg	صبي صغير ينظر إلى رؤوس أخرى	3/5	a group of youths playing	3/5
img_140.jpeg	طفل صغير يلعب على مسار بالقرب من الماء	2/5	a family playing outdoors	2/5
img_141.jpeg	رجل يحمل رأسًا على رأسه	2/5	a man wearing glasses stands in the background	2/5
img_142.jpeg	رجلان يحملان صن على الرصيف	3/5	Two guys are standing on the side of the road	4/5
img_143.jpeg	صبي صغير يلعب على الرصيف	2/5	a young boy is playing on the lawn	2/5
img_144.jpeg	يلعب على العشب الأخضر	2/5	a group of people playing	1/5
img_145.jpeg	رجلين يحمل غطس على رأسه	2/5	a group of people wearing socks and sunglasses are sitting in the water	3/5
img_146.jpeg	رجل ينظر إلى الهاتف المحمول	1/5	a young girl wearing sunglasses and glasses on the corner	3/5
img_147.jpeg	رجل يحمل رؤوس على رأسه	2/5	a group of people playing skateboards	2/5
img_148.jpeg	يلعب على العشب الأخضر	1/5	a very typical rectangle	1/5
img_149.jpeg	فتاة صغيرة تنزل على دراجة أخرى	2/5	A young girl in a purple dress is sitting on the floor	2/5
img_15.jpeg	يحمل طائرًا في الهواء	2/5	a r s across the ear	1/5
img_150.jpeg	رجل يحمل رأسًا على رأسه	2/5	A man in a t shirt is walking	3/5
img_151.jpeg	يلعب على العشب الأخضر	1/5	a girl playing on the corner	2/5
img_152.jpeg	رجلين يحملان حقًا على جبين مغطى بالثلوج	3/5	Two people in a spak on the side of the road	2/5
img_153.jpeg	يلعب على العشب	2/5	a group of people sitting on the floor	1/5
img_154.jpeg	رعاة الرعاية على العشب الأخضر	1/5	Two s are being played by a trophy	2/5

img_155.jpeg	رجلين يرتاد سترات زرقاء يقف على رصيف صغير	1/5	a group of guys playing on the dock	3/5
img_156.jpeg	فتاة على رصيف مرصفة بالحصى	2/5	a girl is sitting on the side of the road	3/5

APPENDIX B: Fine-Tuning a Bilingual Image Captioning Model with CLIP-ViT Embeddings and mT5 using LoRA in PyTorch

```
# ===== INSTALLATION =====
!pip install transformers openpyxl peft

# ===== IMPORTS =====

import os

import torch

import pandas as pd

import torch.nn as nn

import torch.optim as optim

from torch.utils.data import Dataset, DataLoader

from transformers import MT5ForConditionalGeneration, MT5Tokenizer

from peft import get_peft_model, LoraConfig

import matplotlib.pyplot as plt

from tqdm import tqdm

# ===== DEVICE SETUP =====

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# ===== PATHS =====

caption_file = "/kaggle/input/new-model/output_Bilingual_captions_2_new.xlsx"

embedding_dir = "/kaggle/input/clipvit-model/"

checkpoint_path = "/kaggle/input/arabic37/final_arabic_checkpoint_epoch_37.pth"

final_checkpoint_path = "/kaggle/working/final_arabic_checkpoint_epoch_38.pth"

# ===== LOAD DATA =====

captions_df = pd.read_excel(caption_file, engine="openpyxl")

captions_df.columns = ["image_name", "caption"]

# Detect language

def detect_language_token(text):

    return "<ARABIC>" if any('\u0600' <= c <= '\u06FF' for c in text) else "<ENGLISH>"
```



```

captions_df["language"] = captions_df["caption"].apply(detec_t_language_token)
# Filter valid embeddings
captions_df = captions_df[
    captions_df["image_name"].apply(lambda x: os.path.exists(os.path.join(embedding_dir, x
+ ".pt"))))
].reset_index(drop=True)
# ===== DATASET =====
class CaptionDataset(Dataset):
    def __init__(self, df, embedding_dir):
        self.embedding_dir = embedding_dir

        self.data = [(row["image_name"], row["language"], row["caption"]) for _, row in
df.iterrows()]

    def __len__(self):
        return len(self.data)

    def __getitem__(self, idx):
        image_name, language, caption = self.data[idx]

        embedding_path = os.path.join(self.embedding_dir, image_name + ".pt")

        hidden_state = torch.load(embedding_path)

        return hidden_state, language, caption
# ===== DATALOADER =====
dataset = CaptionDataset(captions_df, embedding_dir)
train_loader = DataLoader(dataset, batch_size=8, shuffle=True)
# ===== MODEL =====
class CaptioningModel(nn.Module):
    def __init__(self, mt5_model_name, mt5_hidden_size):
        super().__init__()

        self.tokenizer = MT5Tokenizer.from_pretrained(mt5_model_name)

        special_tokens = {"additional_special_tokens": ["<ENGLISH>", "<ARABIC>"]}

        self.tokenizer.add_special_tokens(special_tokens)

```

```

self.mt5 = MT5ForConditionalGeneration.from_pretrained(mt5_model_name)

self.mt5.resize_token_embeddings(len(self.tokenizer))

lora_config = LoraConfig(
    r=32, lora_alpha=64, target_modules=["q", "v"],
    lora_dropout=0.05, bias="none", task_type="SEQ_2_SEQ_LM"
)

self.mt5 = get_peft_model(self.mt5, lora_config)

self.cls_to_prompt1 = nn.Linear(768, mt5_hidden_size)

self.dropout = nn.Dropout(0.1)

self.cls_to_prompt2 = nn.Linear(mt5_hidden_size, mt5_hidden_size)

def forward(self, languages, pixel_values, captions=None):
    pixel_values = pixel_values.to(device)
    prompt_embed = self.cls_to_prompt1(pixel_values)
    prompt_embed = self.dropout(prompt_embed)
    prompt_embed = self.cls_to_prompt2(prompt_embed)
    if captions:
        inputs = [lang + " " + cap for lang, cap in zip(languages, captions)]
        cap_input_ids = self.tokenizer(
            inputs, return_tensors="pt", padding=True, truncation=True, max_length=64
        ).input_ids.to(device)
        outputs = self.mt5(inputs_embeds=prompt_embed, labels=cap_input_ids)
        return outputs
    else:
        gen_ids = self.mt5.generate(inputs_embeds=prompt_embed, max_length=50)
        return self.tokenizer.batch_decode(gen_ids, skip_special_tokens=True)

# ===== INIT MODEL =====

model = CaptioningModel("google/mt5-base", 768).to(device)

```

```

# ===== TRAINING SETUP =====

learning_rate = 1e-5

optimizer = optim.AdamW(model.parameters(), lr=learning_rate)

clip_value = 1.0

train_losses = []

# ===== LOAD CHECKPOINT =====

start_epoch = 0

if os.path.exists(checkpoint_path):
    checkpoint = torch.load(checkpoint_path, map_location=device)
    model.load_state_dict(checkpoint["model_state_dict"])
    optimizer.load_state_dict(checkpoint["optimizer_state_dict"])
    start_epoch = checkpoint["epoch"]
    train_losses = checkpoint.get("train_losses", [])
    print(f" Loaded checkpoint from epoch {start_epoch + 1}")
else:
    print(" Starting from scratch.")

# ===== TRAINING LOOP =====

epochs = 38

grad_accum_steps = 8

for epoch in range(start_epoch, epochs):
    model.train()
    train_loss = 0
    num_batches = 0
    optimizer.zero_grad()
    progress_bar = tqdm(train_loader, desc=f" Training Epoch {epoch+1}/{epochs}")
    for step, batch in enumerate(progress_bar):
        pixel_values, languages, captions = batch
        pixel_values = pixel_values.to(device)
        optimizer.zero_grad()
    progress_bar = tqdm(train_loader, desc=f" Training Epoch {epoch+1}/{epochs}")
    for step, batch in enumerate(progress_bar):
        pixel_values, languages, captions = batch
        pixel_values = pixel_values.to(device)

```

```

outputs = model(languages, pixel_values, captions)

    loss = outputs.loss / grad_accum_steps

    loss.backward()

    if (step + 1) % grad_accum_steps == 0 or (step + 1) == len(train_loader):

        nn.utils.clip_grad_norm_(model.parameters(), max_norm=clip_value)

        optimizer.step()

        optimizer.zero_grad()

    train_loss += loss.item()

    num_batches += 1

    progress_bar.set_postfix({"Step Loss": loss.item() * grad_accum_steps})

avg_train_loss = train_loss / num_batches if num_batches > 0 else float("inf")

train_losses.append(avg_train_loss)

print(f" Epoch {epoch+1}, Final Average Loss: {avg_train_loss:.4f}")

torch.save({

    "epoch": epoch + 1,

    "model_state_dict": model.state_dict(),

    "optimizer_state_dict": optimizer.state_dict(),

    "train_losses": train_losses

}, final_checkpoint_path)

print(f" Checkpoint saved to {final_checkpoint_path}")

# ===== LOSS VISUALIZATION =====

plt.figure(figsize=(8, 5))

plt.plot(train_losses, marker='o')

plt.title(" Training Loss per Epoch")

plt.xlabel("Epoch")

plt.ylabel("Loss")

plt.grid(True)

plt.tight_layout()

plt.show()

print("\n=== Training Loss per Epoch ===")

for i, loss in enumerate(train_losses, 1):

    print(f"Epoch {i}: Loss = {loss:.4f}")

```

APPENDIX C: Bilingual Caption Gradio Website

```
import torch
import torch.nn as nn
import re
import os
from transformers import MT5Tokenizer, MT5ForConditionalGeneration
from peft import get_peft_model, LoraConfig
import gradio as gr

# ===== CONFIG =====
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

embedding_dir = r"C:\Users\USER\Downloads\vit_embeddings-200"
checkpoint_path = r"C:\Users\USER\Documents\grad_project\final1.pt"

# ===== CLEANING FUNCTION =====
def clean_caption(caption):
    caption = caption.strip()
    caption = re.sub(r"\s+", " ", caption)
    caption = caption.replace("..", ".").replace(",", ",")
    caption = re.sub(r"([?!,,:])\1+", r"\1", caption)
    caption = re.sub(r"\b(\w+)(\1\b)+", r"\1", caption)
    return caption

# ===== MODEL =====
class BilingualCaptioningModel(nn.Module):
    def __init__(self, mt5_model_name, mt5_hidden_size):
        super().__init__()
        self.tokenizer = MT5Tokenizer.from_pretrained(mt5_model_name)
        self.tokenizer.add_special_tokens({"additional_special_tokens":
["<ENGLISH>", "<ARABIC>"]})
        self.mt5 = MT5ForConditionalGeneration.from_pretrained(mt5_model_name)
        self.mt5.resize_token_embeddings(len(self.tokenizer))

        lora_config = LoraConfig(
            r=32, lora_alpha=64, target_modules=["q", "v"],
            lora_dropout=0.05, bias="none", task_type="SEQ_2_SEQ_LM"
        )
        self.mt5 = get_peft_model(self.mt5, lora_config)

        self.cls_to_prompt1 = nn.Linear(768, mt5_hidden_size)
        self.dropout = nn.Dropout(0.1)
        self.cls_to_prompt2 = nn.Linear(mt5_hidden_size, mt5_hidden_size)

    def encode_prompt(self, pixel_values):
        prompt_embed = self.cls_to_prompt1(pixel_values)
        prompt_embed = self.dropout(prompt_embed)
        prompt_embed = self.cls_to_prompt2(prompt_embed)
```

```

return prompt_embed

def generate_caption(self, pixel_values, lang_token):
    decoder_start_token_id = self.tokenizer.convert_tokens_to_ids(lang_token)
    prompt_embed = self.encode_prompt(pixel_values)
    gen_ids = self.mt5.generate(
        inputs_embeds=prompt_embed,
        decoder_start_token_id=decoder_start_token_id,
        max_length=64,
        num_beams=5,
        top_k=50,
        top_p=0.95,
        temperature=1.0,
        no_repeat_ngram_size=2,
        repetition_penalty=1.2,
        length_penalty=1.0,
        early_stopping=True
    )
    decoded = self.tokenizer.batch_decode(gen_ids, skip_special_tokens=True)
    return [clean_caption(c) for c in decoded]

# ===== LOAD MODEL =====
model = BilingualCaptioningModel("google/mt5-base", 768).to(device)
checkpoint = torch.load(checkpoint_path, map_location=device)
if "model_state_dict" in checkpoint:
    model.load_state_dict(checkpoint["model_state_dict"], strict=False)
else:
    model.load_state_dict(checkpoint, strict=False)
model.eval()

# ===== CAPTION FUNCTION =====
def generate_caption_with_name(image, filename, lang_choice):
    if image is None or filename.strip() == "":
        return "⚠ Please upload image and enter image file name without extension."

    base_name = filename.strip()
    pt_path = os.path.join(embedding_dir, base_name + ".pt")

    if not os.path.exists(pt_path):
        return f"⚠ No embedding file found for '{base_name}.pt' in embeddings folder."

    embedding = torch.load(pt_path).unsqueeze(0).to(device).float()

    with torch.no_grad():
        if lang_choice == "Arabic":
            caption = model.generate_caption(embedding, "<ARABIC>")[0]
            return f"SA Arabic: {caption}"

```

```

else:
    caption = model.generate_caption(embedding, "<ENGLISH>")[0]
    return f"GB English: {caption}"

# ===== CUSTOM GREEN THEME
=====
custom_css = """
body {
    background-color: #e6f2e6 !important;
    font-family: 'Segoe UI', sans-serif;
}

h1, h2, h3, label, p {
    color: #1b5e20 !important;
    font-weight: bold;
}

.gr-button {
    background: linear-gradient(135deg, #43a047, #2e7d32) !important;
    color: white !important;
    font-weight: bold !important;
    border-radius: 10px !important;
    padding: 10px 20px !important;
    box-shadow: 0px 3px 6px rgba(0,0,0,0.2) !important;
}

input, textarea, .gr-textbox, .gr-radio, .gr-image {
    border: 2px solid #81c784 !important;
    border-radius: 8px !important;
    background-color: #f1f8e9 !important;
    color: #1b5e20 !important;
}

.gr-textbox textarea {
    font-weight: bold !important;
    font-size: 16px !important;
}
"""

# ===== GRADIO INTERFACE
=====
with gr.Blocks(css=custom_css) as iface:
    gr.Markdown("## 🌐 Bilingual Image Captioning Tool")

    with gr.Row():
        with gr.Column():
            image_input = gr.Image(type="pil", label="📁 Upload Image")
            filename_input = gr.Textbox(label="📄 Enter Image File Name (without extension)", placeholder="e.g. img_184")

```

```
lang_choice = gr.Radio(choices=["Arabic", "English"], value="Arabic",
label="🌐 Select Language")
submit_btn = gr.Button("🔗 Generate Caption")

with gr.Column():
    caption_output = gr.Textbox(label="🖼️ Generated Caption", lines=4)

submit_btn.click(fn=generate_caption_with_name,
                inputs=[image_input, filename_input, lang_choice],
                outputs=caption_output)

iface.launch()
```